

EthernaCare

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[Ethernacare]

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Declaration

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Abstract

EthernaCare is a safety-support mobile application developed to address delayed emergency or death discovery among older adults and other individuals who live alone. This issue is becoming more important as Malaysia moves towards an ageing society and more individuals live separately from their family members. When a person living alone experiences an emergency and is unable to request assistance, the situation may remain unnoticed for a long period. EthernaCare aims to reduce this delay while also helping users prepare and securely communicate their post-death preferences.

The application uses a gamified virtual companion named Oren to encourage regular interaction. Instead of using a fixed daily check-in, each user can configure a rolling inactivity threshold from one to 168 hours. A successful check-in is recorded when the current threshold window has expired. When a user misses the first window, the application displays a local reminder. The second missed window sends an SMS to the user's verified phone number, while the third missed window escalates the alert to the selected trusted contact. Oren also includes energy changes, feeding, playing, virtual tokens, owned toys, a shop and weather-based backgrounds. Check-in streaks separately allow users to collect virtual badges and vouchers.

EthernaCare also provides trusted-contact management, emergency location sharing, AI-assisted general guidance and protected Legacy Planning. The Legacy feature allows users to record funeral preferences, Legacy Notes and private documents. After 90 days without a successful check-in, the owner receives a warning and has 24 hours to check in or cancel the process. Otherwise, the verified primary contact may receive protected access for seven days.

The system was developed using Flutter and Supabase with supporting Edge Functions, scheduled server processing and external SMS, email, weather and AI services.

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Chapter 1

Introduction

1 Introduction

This chapter presents the background and motivation of the Ethernacare project. It explains the social problem related to elderly individuals living alone and the risks linked to delayed discovery during emergencies. The chapter also outlines the project objectives, scope, and significance to provide a clear understanding of what the system aims to achieve.

1.1 Project Objective(s)

The first objective of Ethernacare is to provide an engaging safety check-in mechanism for older adults and other individuals who live alone. Users interact with a virtual companion named Oren to record a successful check-in. Instead of requiring a fixed daily interaction, the system allows the user to configure a rolling inactivity threshold between one and 168 hours. This approach provides flexibility for different user routines while still allowing the system to identify prolonged inactivity.

The second objective is to provide a staged inactivity response process. When the first inactivity window is missed, the application provides a local reminder to the user. When the second window is missed, the system attempts to send an SMS to the user's verified phone number. If the user remains inactive until the third window, the system escalates the SMS notification to the selected trusted contact. This process is intended to reduce unnecessary alerts while still providing an additional safety layer for users who may require assistance.

The third objective is to provide secure Legacy Planning functions. Users can record their funeral preferences, write protected Legacy Notes and upload private documents. The system also provides a server-controlled Legacy release process involving owner consent, a unique Legacy UID, a verified primary contact and protected access verification. If the user remains inactive for 90 days, the system warns the owner and provides 24 hours to check in or cancel the process before temporary access may be released to the verified primary contact.

The fourth objective is to improve regular user engagement through gamification. Oren has an energy level that changes according to inactivity and can be cared for through feeding and playing actions. Users can earn virtual tokens, purchase toys and select owned items for Oren. Check-in streaks separately allow users to unlock virtual badges and vouchers, providing motivation without requiring physical product delivery.

The fifth objective is to provide accessible general information through AI-assisted guidance. The AI feature helps users find basic information related to elderly care, emergency

preparation and funeral arrangements. However, its responses are provided for general informational purposes only and are not medical, legal, emergency or professional advice.

Overall, Ethernacare supports “SDG 3 – Good Health and Well-Being” by providing a practical smartphone-based safety-support application. The system does not diagnose medical conditions, confirm a user’s physical health, guarantee emergency rescue or guarantee successful SMS delivery. Its purpose is to reduce the possibility that prolonged inactivity remains unnoticed and to support communication between users and their trusted contacts.

1.2 Background of the Project

1.2.1 Malaysia is about to become an aging society

According to the prediction from “Portal Rasmi Kementerian Kewangan”, Malaysia will become an aged nation in 2048 (Azizan, 2025). The number of elderly people living alone will definitely increase as the elderly population increases. Moreover, many elderly people have poorer health compared to younger people and face a higher risk of death. These conditions increase the chance of dying alone without immediate discovery.

1.2.2 Migration and Elderly Parents Living Alone

In recent years, many young adults have moved from home to other areas for work or education. Internal migration remained dominant at 63.5% in 2024 (up from 62.3% in 2022), Urban to urban migration dominated at 84.6% in 2024 (up from 79.3% in 2022)(*Migration Survey Report, Malaysia, 2024, 2025*). This often happens when the children are grown up and want to start their own lives. As a result, parents are left living alone in their house. The contact and interaction between parents and children becomes less frequent, children are busy in their own business and usually only back to home on certain holidays. When elderly parents live alone, they are facing the higher risks of dying alone because help may not be available immediately. This situation increases social isolation.

1.2.3 Increase in the Number of Single Individuals

The number of marriages decreased 12.5 per cent from 215,022 (2022) to 188,100 (2023). (*Migration Survey Report, Malaysia, 2024, 2025*). In 2020, there were 14.5 male and 14.3 female births per thousand population, a decrease from 15.1 male and 14.9 female births per thousand population in 2019 (*International Journal Of Academic Research In Business & Social Sciences, 2024*). It was the lowest rate recorded in the country. Combined with these factors, the number of single individuals without children will increase in the future. Against the backdrop of soaring consumption levels, many people are facing delayed marriage, financial issues, work pressure or personal choices. Others remain single due to divorce or the loss of a spouse. As a result, more adults live alone for longer periods of time. This change affects daily support systems, especially as people grow older. Decades later, the number of living alone elderly people will inevitably increase.

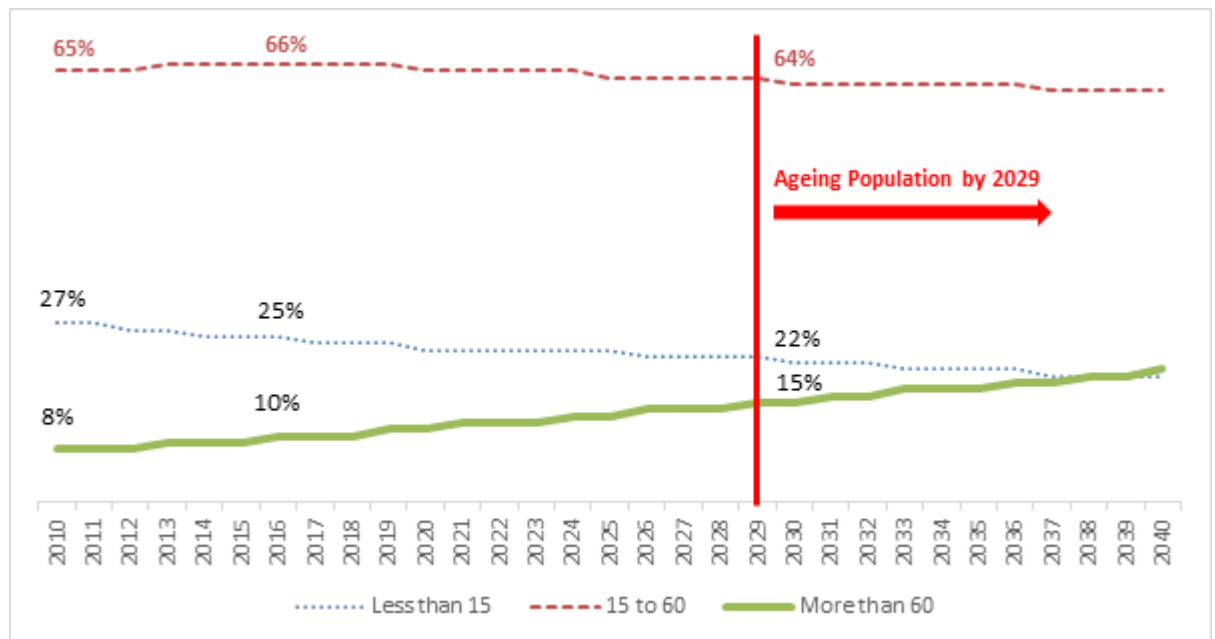


Figure 1.2.1: The prediction of the population age group in the future

(Malaysia Population Research Hub, n.d.)

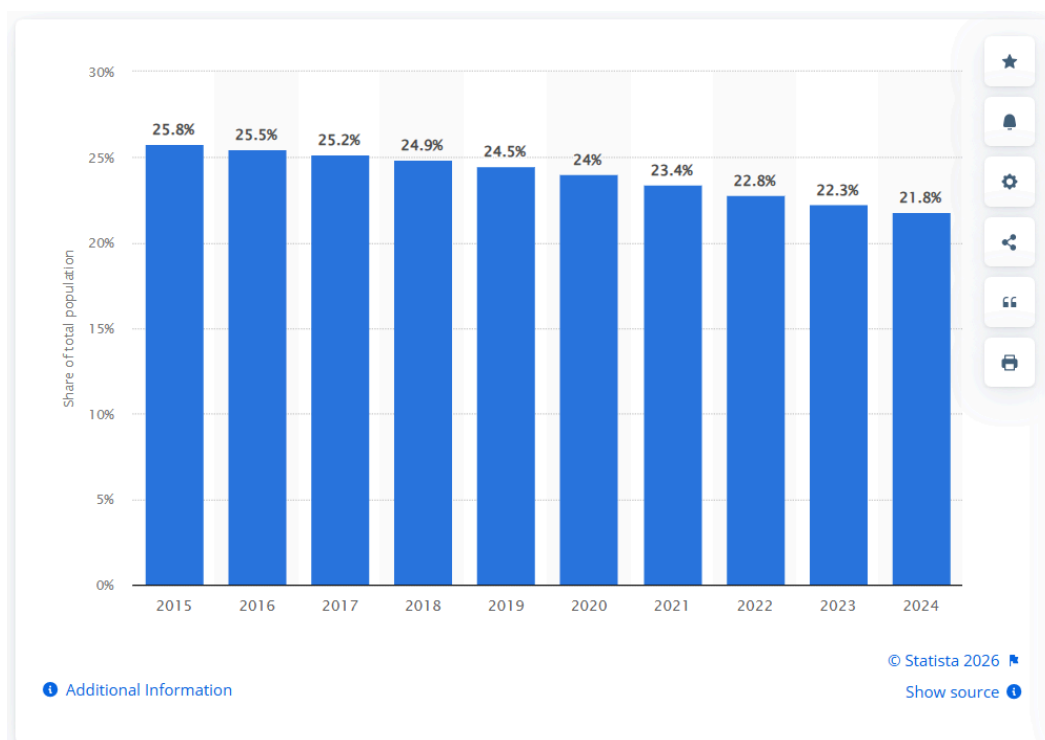


Figure 1.2.2: Children in total population in Malaysia from 2015 to 2024

(Statista, 2025)

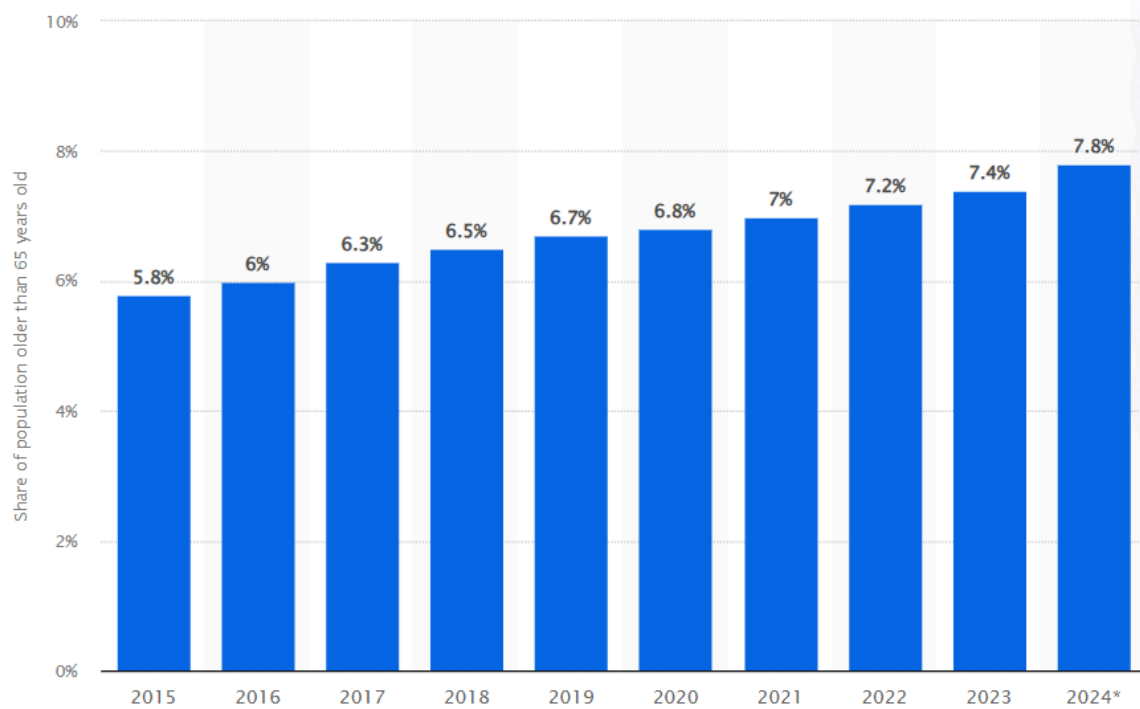


Figure 1.2.2: Aging population Malaysia 2015-2024

(Statista, 2025)

1.3 Advantages & Contributions

1.3.1 Easy to access and achieve

The project provides a solution that does not require extra devices or complicated devices. It relies on the smartphones that are owned by the users. The system uses a **gamified pet-care interaction** that encourages regular use. By transforming the check-in into the act of 'caring' for a virtual cat, the system reduces the 'surveillance' feel of the app and replaces it with a sense of companionship and purpose.

1.3.2 Reducing the unfortunate cases & Notified families in time

EthernaCare contributes by providing a structured process for identifying prolonged user inactivity and attempting to notify the appropriate person. When a configured inactivity window is missed, the application first reminds the user locally. Continued inactivity causes the system to attempt an SMS to the user's verified phone number before escalating to the selected trusted contact. When location permission is granted and a valid device location is available, a Google Maps link may also be included in the emergency message. This process may help family members or trusted contacts respond earlier to a possible emergency or delayed death discovery. However, successful notification depends on internet connectivity, external service availability, mobile carrier delivery and correct contact information.

1.4 Project Plan

The project follows an Agile mobile application development approach. Development is divided into several phases that include planning, system design, implementation, testing, and deployment. Core features such as activity detection and emergency alert notification are developed early in the project. Features related to funeral preferences, will management, and AI-assisted guidance are developed in later stages. Testing is carried out throughout the development process to ensure that each function works correctly and meets user needs.

Project Schedule:		
ACTIVITIES	EXPECTED OUTCOME	COMPLETION DATE
Proposal	Determine the objective, problems and solutions.	16/12/2025
Phase 2: FYP1		
Chapter 1: Introduction	Introduce the project purpose, background, objective and justification	26/1/2026
Chapter 2: Research background	Retrieve the relevant information, system requirements and research gaps	7/2/2026
Chapter 3: Methodology	Defined the system development tools and requirements.	24/2/2026
Chapter 4: System Design	Mock up prototype, system architecture and data flow diagram	14/3/2026
Project I Portfolio	Submission	13/4/2026
Phase 3: FYP2		
Chapter 5: Development & Testing	Develop the main functions	26/6/2026
Chapter 6: System deployment	Deploy and test the system in real-world scenarios.	6/7/2026
Chapter 7: Project conclusion	Final summary and outcome of the project	1/8/2026

Figure 1.4: The project schedule for each phases

1.5 Project Team & Organization

This project is developed as an individual final year project under the guidance of a project supervisor. The student is responsible for system design, development, testing, and documentation. The supervisor provides academic guidance, technical advice, and feedback to ensure that the project meets academic requirements and follows proper development practices.

1.6 Chapter Summary and Evaluation

This chapter introduced Ethernacare and explained the social problem associated with older adults and other individuals living alone. The system uses Oren, a virtual companion, to make safety check-ins more engaging and less clinical. Users can configure a rolling inactivity threshold from one to 168 hours, while staged reminders and SMS escalation help identify prolonged inactivity. In addition to check-in monitoring, Ethernacare provides Oren energy management, tokens, toys, virtual rewards, trusted contacts, emergency location support, AI-assisted guidance and protected Legacy Planning. The project is designed as a safety-support tool and does not replace medical monitoring, emergency services or direct family communication.

Chapter 2

Literature Review

2 Literature Review

This chapter examines previous research and existing systems related to elderly individuals who live alone and the growing concern of lonely deaths. It looks at demographic changes such as population ageing and the rise in independent living among older adults. It also reviews current technology solutions, including wearable emergency devices, smart home monitoring systems, and mobile safety applications. Through this review, the chapter highlights the strengths and weaknesses of existing approaches and clarifies how the proposed project addresses identified gaps and limitations.

2.1 Aging Population in Malaysia

2.1.1 Demographic Shift Toward an Ageing Society

Malaysia is moving toward an ageing nation as the percentage of citizens aged 60 and above continues to increase. The Department of Statistics Malaysia (*Current Population Estimates, Malaysia, 2023, 2023*) projects that older adults will make up around **15%** of the population by 2030. This change is mainly influenced by longer life expectancy and lower birth rates. The transition creates new challenges in healthcare planning, elderly care services, and community support systems.

2.1.2 Increase in Elderly Living Alone

The increase in elderly individuals living alone is closely linked to broader demographic and social transitions. According to the *World Report on Ageing and Health* by the World Health Organization (2015), although more generations within a family may now be alive at the same time due to increased life expectancy, these generations are increasingly living separately. In many countries, the proportion of older people living alone is rising dramatically. For example, in several European countries, more than 40% of women aged 65 and above live alone. Even in societies with strong traditions of multigenerational living, such as Japan and India, extended family households are becoming less common. Declining fertility rates mean there are fewer younger family members available to provide care, while changing gender roles have reduced the traditional caregiving capacity within families. Smaller household sizes may limit reciprocal care arrangements and increase the risk of poverty. Furthermore, older adults living alone may face higher risks of social isolation and suicide. Although many older people prefer to remain in their own homes and communities for as long as possible,

these structural changes suggest that traditional family-based care models are becoming less sustainable in the modern era (World Health Organization, 2015).

2.2 Technology concepts

2.2.1 Smartphone-Based Passive Monitoring

Smartphone-based monitoring systems use built-in sensors and activity logs to observe user behavior without requiring constant manual input (Wang, 2014). This passive approach allows the system to detect unusual patterns such as prolonged inactivity, which may indicate potential emergencies. Because the monitoring process runs automatically in the background, it reduces the need for continuous interaction and can operate without disturbing users.

2.2.2 Daily Check-In Monitoring Systems

Daily check-in monitoring systems are used in safety and well-being applications to verify that users are active through simple interactions with a mobile device. Instead of continuously collecting sensor data, gamified check-in systems utilize **virtual pets or interactive elements** to maintain user engagement. By requiring a tactile interaction like 'petting' an on-screen character, the system ensures the user is physically capable and responsive, serving as a more interactive form of passive monitoring (Rashidi & Mihailidis, 2013).

2.2.3 Wearable or Sensor Monitoring

Wearable and sensor-based monitoring systems are commonly used in elderly care technologies to continuously track health conditions and daily activities. These systems typically involve wearable devices such as smartwatches or sensor bands that collect physiological and movement data, including heart rate, walking patterns, and sleep activity. The collected data can be analyzed to identify abnormal patterns such as falls, prolonged inactivity, or sudden changes in physical behaviour (Patel et al., 2012).

2.2.4 Duolingo and Behavioral Gamification

Modern gamification success stories, most notably **Duolingo**, demonstrate that habit formation is driven by emotional engagement and loss aversion. According to research on Duolingo's growth strategy, the use of a persistent mascot (Duo the Owl) and a daily "streak" mechanic transforms a high-effort task into a "sticky" daily routine (Hasan, 2026). By quantifying consistency, users feel a sense of ownership over their progress—a psychological

phenomenon known as the **Endowment Effect**. Carefully designed behavioral incentives can also support health-related behavior change (Vlaev et al., 2019).

EthernaCare adapts these proven tactics by replacing standard check-in prompts with a virtual cat companion. By leveraging the emotional bond between the user and the digital pet, the system ensures high retention rates, as users are motivated to "check-in" to maintain their pet's well-being and their own reward progress.

2.3 Technology Acceptance Among the Elderly

2.3.1 Smartphone Accessibility Among Elderly Users

In recent years, smartphone ownership among older adults has increased significantly. According to the Pew Research Center (2021), smartphone adoption among individuals aged 65 and above has grown substantially compared to previous decades. The increased accessibility of smartphones provides opportunities to develop mobile-based solutions for elderly safety and monitoring. Since many elderly individuals already own smartphones, mobile applications can serve as cost-effective tools without requiring additional hardware (Pew Research Center, 2025).

2.3.2 Perceived Usefulness and Ease of Use

Recent studies show that perceived usefulness and ease of use remain strong predictors of technology adoption among older adults. Research on digital health and mobile monitoring systems indicates that elderly users are more likely to adopt applications when they believe the system improves their safety and supports independent living (Anderson et al., 2020). Confidence in smartphone usage and prior experience with digital tools also influence acceptance. When applications are perceived as complicated or unnecessary, adoption rates decrease.

2.3.3 Importance of User-Friendly Design

Usability plays a critical role in elderly technology adoption. Studies on mobile health applications emphasize that simple navigation, readable fonts, clear instructions, and minimal interaction steps significantly improve user satisfaction and long-term engagement (Dusseljee-Peute et al., 2018). Complex menus and small interface elements may discourage elderly users. Therefore, systems designed for older adults should prioritize clarity, accessibility, and reduced cognitive effort.

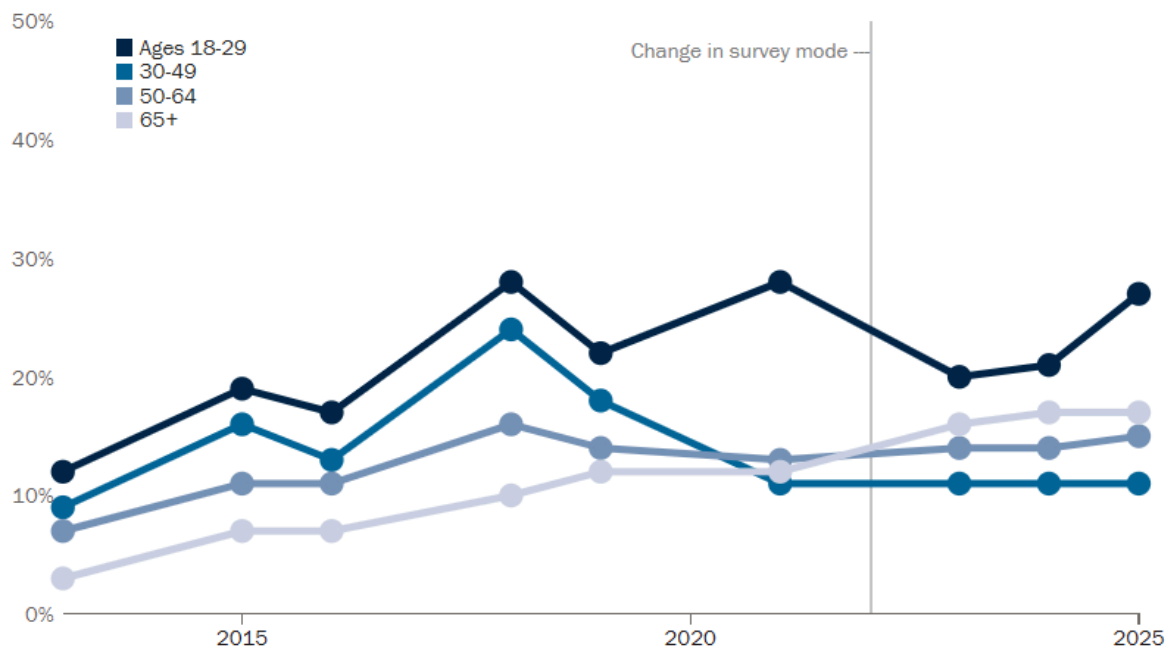


Figure 2.1: Percentages of U.S. adults who are smartphone dependent, by age

(Pew Research Center, 2025)

2.4 Similar Existing Systems in the market

2.4.1 Apple Emergency SOS

Apple Emergency SOS is a built-in smartphone feature that allows users to quickly contact emergency services and notify selected emergency contacts. The system can be activated using hardware button combinations, and it automatically shares the user's location with emergency responders and trusted contacts (Apple Inc., n.d.). The main feature is immediate emergency calling with real-time location transmission. It is integrated into iOS devices and depends on manual activation by the user.

This system is related to Ethernacare because both involve emergency alerts and contact notification. The difference is that Ethernacare uses a **daily greeting check-in mechanism** to confirm the user's status instead of requiring manual triggering. In addition, Ethernacare integrates post-death planning and digital document storage, which are not included in Apple Emergency SOS.

2.4.2 CarePredict

CarePredict is a wearable-based monitoring system designed for elderly individuals, particularly those in assisted living facilities. The system uses a wearable device equipped with sensors to monitor daily activities such as eating, sleeping, walking, and bathroom visits. It applies artificial intelligence and behavior pattern analysis to detect anomalies that may indicate health or safety risks (CarePredict, n.d.).

The main feature is continuous activity tracking through wearable sensors combined with AI-based behavioral analytics. The strength of the system lies in its detailed monitoring capabilities. However, it requires users to wear a dedicated device and is often implemented in institutional care settings.

Ethernacare shares the concept of behavior monitoring but differs by using smartphone-based passive detection rather than a wearable device. It is also designed for elderly individuals living independently at home rather than those in assisted living environments.

2.5 Limitation of the project

2.5.1 No Direct Medical Monitoring

Ethernacare does not measure heart rate, blood pressure, oxygen level, body temperature or other medical information. It does not include automatic fall detection and cannot diagnose illnesses or determine the cause of a user's inactivity. A successful Oren interaction only confirms that a check-in was recorded by the application. Therefore, Ethernacare must not be treated as a medical monitoring or diagnostic device.

2.5.2 No Automatic Emergency-Service Dispatch

The application does not automatically contact hospitals, ambulances, police or other Malaysian emergency services. Its automated escalation process attempts to send messages to the verified user and the selected trusted contact. The user may manually contact emergency services when necessary, while the trusted contact remains responsible for deciding what further action should be taken.

2.5.3 External Provider and Internet Dependency

Several functions depend on internet connectivity and third-party providers. SMS delivery depends on the SMS provider and the receiving mobile carrier, while email delivery depends on the email provider and recipient server. AI guidance, server-side processing and cloud database functions also require internet access. Therefore, successful request processing does not always prove that a message was received by its intended recipient.

2.6 Chapter Summary and Evaluation

This chapter analyzed the challenges of elderly isolation and the risks of delayed emergency response. It reviewed technology concepts including passive monitoring and daily check-ins, concluding that **gamification elements**—specifically interactive virtual companions—can bridge the gap between technical safety requirements and user acceptance. This review justifies the implementation of a tactile "petting" interaction as a reliable and less intrusive method of monitoring.

Chapter 3

Methodology and Requirements Analysis

3 Methodology and Requirements Analysis

This chapter describes the methodology and requirements analysis for the proposed system. It explains the development approach used in the project and identifies the stakeholders involved in the system. The chapter also presents the functional and non-functional requirements needed to support the system features. In addition, system models such as the use case diagram are introduced to illustrate how users interact with the system and how the main system functions are organized.

3.1 Methodology

3.1.1 Methodology Used: Agile

The project adopts the Agile methodology, which supports iterative development, continuous evaluation and incremental improvement. Instead of developing the whole application in a single stage, Ethernacare was divided into smaller functional areas, including authentication, Oren care, rolling check-ins, emergency escalation, trusted contacts, virtual rewards, Legacy Planning, AI guidance and weather integration. Each area could be designed, implemented and tested separately before being integrated with the rest of the application.

3.1.2 Methodology Phases

1. **Requirement Gathering** – The first phase identified the functional and non-functional requirements of Ethernacare, including configurable inactivity thresholds, Oren interaction, staged SMS escalation, verified contacts, virtual rewards, protected Legacy access, authentication, biometric session unlock and AI guidance.
2. **System Design** – The application architecture, database model, user interface, activity flows and security controls were designed. The design also identified the responsibilities of Flutter, Supabase, Android WorkManager, Edge Functions, scheduled processing and external providers.
3. **Development** – The application was implemented module by module using Flutter and Dart. Supabase was used for authentication, PostgreSQL data storage, Row Level Security, private file storage, Realtime updates, Edge Functions and scheduled server processing.
4. **Testing** – Automated unit, service, repository and widget tests were conducted to verify deterministic business rules and user-interface behaviour. Static analysis and platform builds were also performed. Functions involving real SMS receipt, GPS accuracy, biometric hardware and exact Android scheduling require separate physical acceptance testing.

5. **Evaluation and Improvement** – Test results and application behaviour were reviewed to identify problems. Requirements, validation rules, interfaces and server functions were refined through repeated Agile iterations until the main application features operated consistently.

3.1.3 Justification for Methodology

The Agile methodology is suitable for this project because the system consists of multiple independent modules, including activity monitoring, emergency alerts, AI chatbot guidance, and document storage. These modules can be developed and tested separately in iterative cycles. Additionally, system requirements may evolve after usability testing with potential users, especially elderly individuals, making a flexible development approach necessary. Iterative development helps reduce project risks by identifying issues early and allowing continuous improvements. It also enables critical features, such as emergency alert functions, to be tested and refined at an early stage of development.

3.2 Requirements Analysis

3.2.1 Stakeholders

1. **Elderly user (Primary)**

The elderly user (primary user) is the main user of the system. This user interacts with the application for passive activity monitoring and performs daily check-ins to indicate that they are safe. The elderly user can also set inactivity time thresholds, register emergency contacts, store funeral preferences and digital will documents, and interact with the AI chatbot for guidance and information. The primary interest of the elderly user is to ensure personal safety and receive faster emergency responses when necessary. In addition, the system allows them to record their personal wishes and ensures that the application remains easy to use and reliable.

2. **Family Members / Trusted Contacts**

Family members or trusted contacts act as emergency recipients and secondary users of the system. They receive alerts when prolonged inactivity is detected or when an emergency situation occurs. During such events, they can access the user's shared location information to

help locate the individual quickly. If permitted by the user, they may also view funeral preferences or authorized documents stored within the system. Their main interests include receiving timely notifications during emergencies, obtaining accurate location information, and understanding the user's final wishes clearly.

3. Independent Individuals Living Alone

Independent individuals living alone are also considered primary users of the system, although they may not necessarily be elderly. These users utilize the same core functions as elderly users, such as activity monitoring, daily check-ins, emergency alerts, and contact notifications. The system provides an additional safety layer for individuals who live independently and may not have someone nearby to check on them regularly. Their main interests include maintaining personal safety while living alone, enabling automatic emergency detection in case of accidents or health issues, and ensuring that trusted contacts or emergency services can be notified quickly during critical situations. This provides them with peace of mind knowing that assistance can be requested if something unexpected happens.

4. System Administrator / Developer

The system administrator or developer acts as the technical stakeholder responsible for maintaining the overall performance and reliability of the system. Their responsibilities include managing system operations, maintaining the database infrastructure, ensuring data security and privacy protection, and performing system updates and bug fixes when necessary. The administrator ensures that the monitoring and emergency alert functions operate correctly and that user data remains protected. Their primary interest is to ensure system stability, maintain security compliance, and guarantee that the application continues to function effectively for all users.

3.2.2 Functional Requirements

Module 1: Gamified Wellness Check Module (Virtual Pet)

One major problem faced by elderly individuals living alone is the risk of **not being noticed when something happens to them**, such as illness, falls, or sudden health emergencies. Studies show that older adults who live alone may remain **unnoticed for long periods when they experience accidents or health problems**, especially if there is no regular communication with family members. For example, research on elderly safety reported that seniors who fall at home sometimes remain on the ground for hours before being rescued because no one is aware of the incident (Melchiorre et al., 2022). Therefore, a daily gamified interaction where the user 'pets' a virtual cat acts as the 'heartbeat signal.' This specific action confirms the user is active and conscious. The system tracks the 'well-being' of the virtual cat, which is directly tied to the user's check-in frequency; if the cat is not 'petted,' it signals a potential emergency for the user.

Module 2: Inactivity and Emergency Response Module

When elderly individuals fail to respond to daily communication or monitoring systems, it may indicate a **potential emergency situation** such as a fall, medical problem, or sudden unconsciousness. Research indicates that elderly people living alone have a **higher risk of falls and delayed rescue**, which can significantly increase health risks or mortality if assistance is not provided quickly (Melchiorre et al., 2022). Therefore, an automated emergency alert system that notifies family members or trusted contacts when the user fails to respond can help ensure that help is requested promptly. This module provides a safety mechanism that reduces response time during emergencies.

Module 3: Contact & Location Management Module

According to research by **Cheng-Wen Lee** and **Hsiu-Mang Chuang**, elderly care platforms commonly include **location tracking, fall detection, and emergency warning services** to improve safety monitoring. Their system uses IoT devices and GPS technology to monitor the location and behavior of elderly patients and notify caregivers when abnormal events occur. This demonstrates that location tracking is an important function in elderly monitoring systems because it enables caregivers to quickly identify the user's location during emergencies (Lee & Chuang, 2016).

Module 4: Virtual Reward Module

Maintaining consistent interaction with monitoring applications can be challenging, particularly for elderly users who may forget or lack motivation to use digital technologies regularly. Behavioral economics research suggests that incentives can play an important role in influencing human behavior. According to a review by Ivo Vlaev and colleagues, financial incentives are widely used in both public and private sectors to encourage desirable behaviors, particularly in health-related programs. Incentive mechanisms can motivate individuals to adopt and maintain positive behavioral changes when appropriately designed (Vlaev et al., 2019).

Based on this concept, the proposed system incorporates a rewarding system to encourage users to perform daily check-ins. When users successfully complete check-ins for a certain number of consecutive days, they may receive small rewards such as virtual voucher or badges. These incentives aim to increase user engagement and motivate consistent interaction with the application, ensuring that the monitoring functions remain effective.

Module 5: AI Assistance Module

Many people are unfamiliar with procedures related to elderly care, emergency arrangements, or funeral planning. Searching for this information online can be confusing and time-consuming, especially during stressful situations. AI-based assistance systems can help users obtain relevant information quickly through conversational interaction. Research on intelligent care systems shows that technology-assisted solutions can support elderly individuals and caregivers by providing accessible information and guidance when needed. Therefore, integrating an AI chatbot can help users and family members quickly obtain guidance and reduce uncertainty during difficult situations (Wang et al., 2025).

Module 6: Legacy Planning Module

The Legacy Planning Module allows users to record funeral preferences, create protected Legacy Notes and store private documents. Research on digital legacy planning shows that digital material can support personal reflection, preserve important information and help individuals communicate their preferences to people who are important to them. However, access, consent, privacy and the management of digital information after death must be carefully considered (Stanley et al., 2026).

The Ethernacare user must deliberately enable Legacy processing and configure a primary contact with a verified phone number and valid email address. A server-controlled process checks the user's Legacy heartbeat. When 90 days have passed without a successful check-in, the owner receives a warning email and is given 24 hours to check in or use a one-time cancellation link. If the owner does not complete a protection action, the primary contact may be notified and a seven-day protected-access window is opened.

The primary contact must provide the correct Legacy UID, a matching phone number and a valid six-digit SMS verification code before protected information is displayed. Legacy Notes and documents are accessible only within the valid release window. A new owner check-in, cancellation, eligibility change or access expiry revokes the protected access.

Private documents should not be permanently exposed through public links. Supabase supports private Storage access through Row Level Security and time-limited signed URLs, which are suitable for temporarily sharing protected files with authorised users (Supabase, 2026).

Module 7: Weather Module

The Weather Module displays regional weather information and weather-based pixel backgrounds according to the Malaysian region selected by the user. The application first requests information from the Malaysian Meteorological Department, which provides general weather forecast data through its Web Service API (Malaysian Meteorological Department, 2026).

When the preferred source is unavailable, the application may use Open-Meteo as a fallback provider. Open-Meteo generates forecast responses using geographical coordinates and combines information from several national weather models (Open-Meteo, 2026). Recent weather results are cached to reduce repeated network requests and improve application responsiveness.

Module 8: Reward Admin Module

The Admin Module provides a separate interface for authorised Ethernacare reward administrators. It is not displayed in the normal user navigation and can only be accessed by an authenticated account registered as active in the `reward_admins` table. Supabase Authentication verifies the administrator's identity, while a protected server function confirms their administrator status before displaying the reward catalogue.

The module allows administrators to create and manage the virtual badges and vouchers available to users. For each reward, the administrator can configure the reward code, title, description, required check-in milestone, reward type, voucher value, and active status. Reward codes and milestone requirements are validated by the server before being saved.

The reward catalogue uses Supabase Realtime to display database changes, and a manual refresh function is also available. Administrators may edit, activate, or deactivate existing rewards. Selected rewards can only be permanently deleted when they have not already been earned by a user. Rewards that have already been earned must be deactivated to preserve historical user records.

Voucher redeem codes are not entered by the administrator. When a user reaches the required milestone, the server automatically awards the reward and generates a unique redeem code for voucher-type rewards. The current Admin Module manages only the virtual reward catalogue and does not process physical deliveries or user reward requests.

3.2.3 Functions not included

1. Funeral Service Booking and Payment System

While the application does not include a marketplace or transaction system. Users cannot browse specific funeral packages, service providers, or make payments for funeral services directly through the app.

2. Direct Communication with Service Providers

The application does not include a real-time chat or messaging system to connect users with funeral directors, lawyers, or religious organizations. The AI chatbot provides general information only and does not facilitate external business communication.

3. Medical and Biological Health Monitoring

The system relies solely on phone activity (screen interaction and app usage). It does not integrate with wearable devices (like smartwatches) to monitor heart rate, falls, or other biological vital signs. It is not a medical diagnostic tool.

4. Legal Will Drafting and Notarization

The application does not provide the legal tools to create, draft, or notarize a will. Users must obtain a legally binding will through traditional legal channels before uploading a copy to the app.

5. Direct Integration with Emergency Dispatch (999/911)

The system is designed to alert trusted contacts and family members first. It does not automatically dial or trigger a direct dispatch from government emergency services (Police or Ambulance) to prevent false alarms that could lead to legal liabilities or resource wastage. Except the user triggers the emergency call manually else the emergency services won't trigger.

3.2.4 Non-Functional Requirements

1. Reliability

- The system shall operate 24/7 without unexpected crashes during background execution.
- Emergency alerts shall include a retry mechanism if the first notification attempt fails.
- User data (contacts, preferences, documents) shall not be lost during normal app updates.
- The system shall automatically resume monitoring after device restart.

2. Usability

- The application interface shall use clear fonts and simple navigation suitable for elderly users.
- Users shall be able to configure inactivity time thresholds easily.
- The emergency status shall be clearly displayed on the home screen.
- The system shall provide confirmation prompts before triggering alerts to reduce false alarms.

3. Performance Efficiency

- The system shall run daily reminder and check-in tracking continuously without causing noticeable device lag.
- The system shall optimize battery usage to prevent excessive power consumption during background monitoring.

3.3 Use Case Diagram

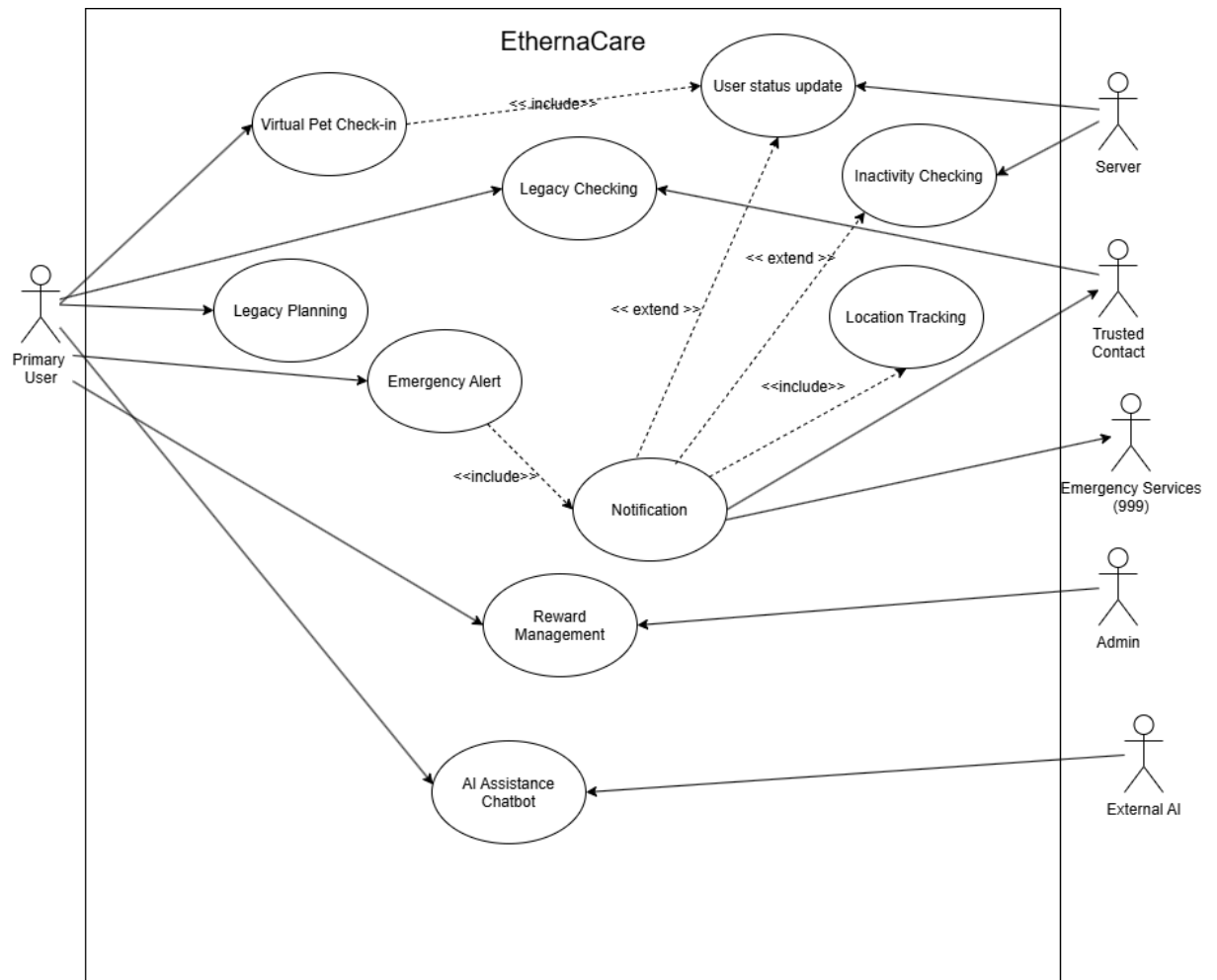


Figure 3.1: Use Case Diagram for each of stakeholders

3.3.1 Use Case Actors**1. Primary User**

- Represents the elderly individual or independent person living alone who uses the application.
- Interacts with the system to perform daily check-ins, trigger emergency alerts, and communicate with the AI chatbot.

2. Trusted Contact

- Receives emergency notifications when the system detects inactivity or when an emergency alert is triggered.
- Can access the user's location information during emergencies.

3. Emergency Services (999)

- Represents emergency services that may be contacted when an emergency call is triggered.

4. External AI

- Provides the artificial intelligence service used by the AI assistance chatbot.

3.3.2 Main Use Case

1. Virtual Pet Interaction (Check-In)

- The user taps virtual pet aka Oren to perform a safety check-in when the configured rolling inactivity threshold has elapsed. A successful check-in records a new heartbeat, resets the inactivity cycle, and updates the user's safety status.
- Feeding, playing with, or petting Oren does not automatically count as a safety check-in unless the check-in action is due and completed.

2. User Status Update

- Updates whether the user's latest check-in is current or overdue according to the configured inactivity threshold.

3. Inactivity Checking

- The scheduled server worker evaluates the latest check-in against the user's configured threshold.

4. Reward Management

- Allows users to earn and collect virtual badges or vouchers after reaching eligible check-in goals. Earned vouchers may contain a user-specific redemption code.

5. Emergency Alert

- A manual SOS alert is activated when the user selects Send Emergency Alert.
- An automatic inactivity alert is separately created by the server after the third missed threshold window.
- The alert records the event and attempts to notify the verified primary trusted contact.

6. Capture Emergency Location

- When the user confirms a manual SOS, Ethernacare attempts to retrieve the device's current location.

7. Notify Primary Trusted Contact

- Sends the appropriate safety alert to the user's verified primary trusted contact.

8. Legacy Planning

- Allows the Primary User to save funeral preferences, select an authorised existing contact, create protected Legacy Notes, and upload supported private documents.

9. Legacy Checking

- Allows the verified primary trusted contact to request access using the user's Legacy UID, registered primary-contact phone, and SMS verification code.
- Funeral preferences are available after successful verification. Protected notes and documents require the 90-day inactivity process, 24-hour owner-protection period, and active seven-day access window.

10. AI Assistance Chatbot

- Allows the user to request general information and guidance. Queries are processed through an external AI service.

11. Notification

- Including sending the reminders message and info trusted contact the user status.

3.4 Chapter Summary and Evaluation

This chapter detailed the Agile methodology and functional requirements for Ethernacare. Key requirements were identified, focusing on the **cat-petting check-in module** as the primary safety verification mechanism. The use case diagrams illustrate how this gamified interaction integrates with the **existing rewarding system**, where consistent "pet care" (check-ins) allows users to earn physical daily necessities. These requirements form the foundation for a system that balances safety, engagement, and practical incentives.

Chapter 4

System Design

4 System Design

This chapter presents the system design of the Ethernacare application. It illustrates how the system is structured and how its components interact to fulfill the required functionalities. The chapter includes the architecture design, class diagram, entity relationship diagram (ERD), and activity diagrams to provide a clear representation of the system's structure and behavior. These design models serve as a blueprint for the system implementation and ensure that the application is well-organized, scalable, and aligned with the defined requirements.

4.1 Architecture Diagram

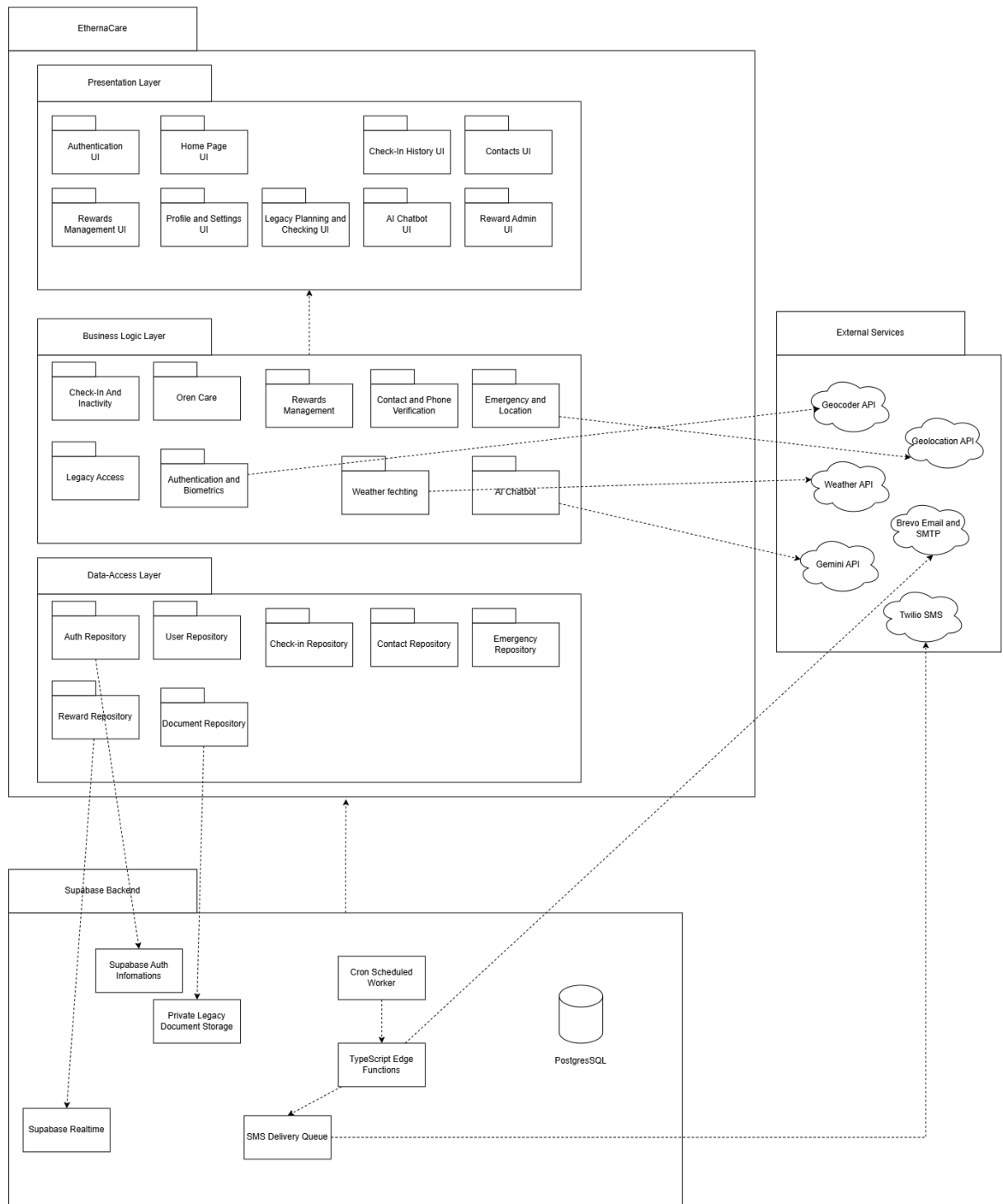


Figure 4.1: 3 Layer + 2 Tier Architecture diagram

4.1.1 Architecture Justification

EthernaCare adopts a logical three-tier architecture to separate the user interface, application processing, and data-access responsibilities. The architecture consists of the **Presentation Layer**, **Business Logic Layer**, and **Data-Access Layer**, supported by a Supabase backend and several external services.

This design improves maintainability, security, testability, and flexibility by giving each layer a defined responsibility. Sensitive and time-dependent operations, including inactivity monitoring, legacy-access processing, document protection, and notification delivery, are handled by server-side Supabase components instead of depending entirely on the user's device.

Presentation Layer

The Presentation Layer manages the interfaces through which users and administrators interact with Ethernacare. It includes:

1. Authentication UI
2. Home Page and Oren UI
3. Check-In History UI
4. Contacts UI
5. Rewards Management UI
6. Profile and Settings UI
7. Legacy Planning and Legacy Checking UI
8. AI Chatbot UI
9. Separate Reward Administration UI

The interfaces are developed using Flutter and are designed to provide clear navigation, readable content, consistent controls, and accessible interaction for elderly users.

This layer collects user input and displays information returned by the Business Logic Layer. It does not directly implement server-side inactivity, notification, or data-protection rules. Separating the interface from these processes allows the visual design to be improved without unnecessarily modifying database operations or server-side processing.

Business Logic Layer

The Business Logic Layer coordinates Ethernacare's application rules and feature workflows. Its main components include:

1. Check-In and Inactivity Management
2. Oren Care and Interaction
3. Reward Management
4. Contact and Phone Verification
5. Emergency and Location Management
6. Legacy Access Management
7. Authentication and Biometric Protection
8. Weather Information Retrieval
9. AI Chatbot Assistance

When a user interacts with the Presentation Layer, the appropriate service validates the input, applies the required rules, and communicates with a repository or Supabase Edge Function.

Security-sensitive and time-dependent rules are also enforced by the backend. For example, Supabase scheduled workers evaluate inactivity thresholds even when the application is closed. Server-side functions also process legacy-access periods, SMS queues, email notifications, phone verification, emergency alerts, and protected document operations.

This arrangement prevents important safety processes from relying solely on the user's phone.

External Services

Ethernacare integrates with external services through controlled service classes or Supabase Edge Functions:

1. **Platform geocoder:** Validates the user's entered home address.
2. **Device geolocation:** Retrieves the current position when the user confirms an SOS alert and grants location permission. Ethernacare does not continuously track the user.
3. **Weather APIs:** Retrieve weather information from data.gov.my and Open-Meteo.
4. **Brevo Email and SMTP:** Delivers authentication emails and transactional legacy-access notifications.
5. **Twilio SMS:** Delivers phone verification codes and configured server-generated SMS alerts.
6. **External AI provider:** Supports Gemini for AI Guidance.

System Data Flow

1. The user interacts with an Ethernacare screen.
2. The Presentation Layer passes the request to a service in the Business Logic Layer.
3. The service validates the input and applies the relevant application rules.
4. A repository sends the authenticated request to Supabase, or the service invokes an Edge Function.
5. Supabase Authentication and Row Level Security validate access.
6. The backend reads or updates PostgreSQL, Storage, Realtime or a notification queue.
7. An Edge Function contacts an external provider when email, SMS or AI processing is required.
8. The result is returned to the Flutter application and displayed to the user.

Data Access Layer

The Data-Access Layer manages communication between the Business Logic Layer and the Supabase backend. It contains the following repositories:

1. Authentication Repository
2. User Repository
3. Check-In Repository
4. Contact Repository
5. Emergency Repository
6. Reward Repository
7. Document Repository

The repositories perform structured operations through the Supabase client, including authenticated table queries, Remote Procedure Calls, private file-storage requests, Edge Function invocation, and Realtime subscriptions.

4.1.2 Architecture Advantages

- **Maintainability:** Interface, application and data-access responsibilities are organised into separate components, making defects and future changes easier to manage.
- **Security:** Authentication, Row Level Security, private storage and server-side functions prevent clients from directly performing privileged operations.
- **Reliability:** Scheduled inactivity and Legacy Access processing continue on the server even when the user's phone is offline or the application is closed.
- **Scalability:** Supabase backend components and external integrations can scale separately from the installed Flutter application.
- **Flexibility:** Repository and service abstractions allow components or external providers to be changed with limited effects on other parts of the system.
- **Testability:** Services, repositories, widgets and server functions can be tested according to their individual responsibilities.

(GeeksforGeeks, 2025)

4.2 Activity Diagram

4.2.1 SOS Emergency Activity

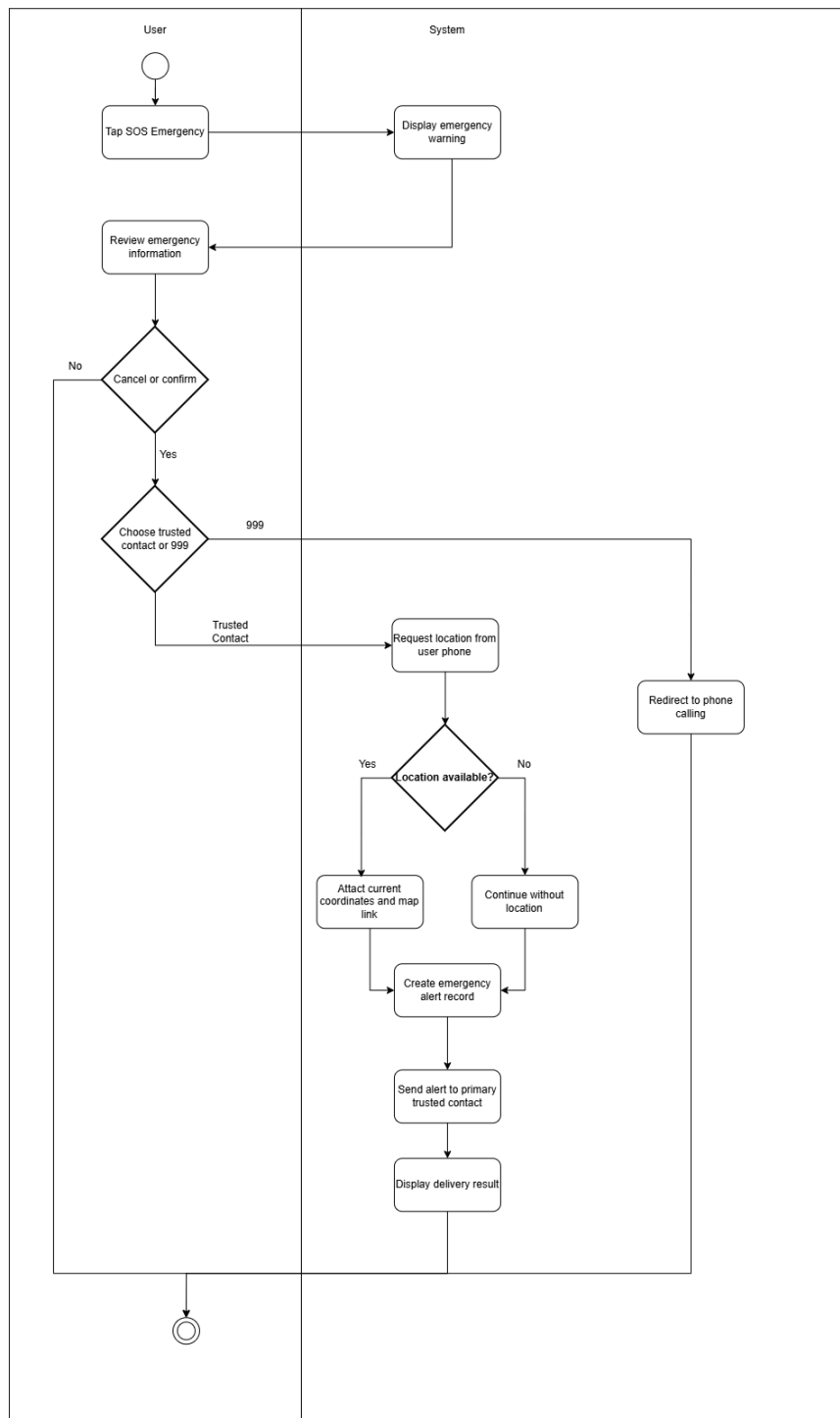


Figure 4.2.1: Activity Diagram show the Emergency Flow of the Ethernacare

The SOS emergency activity begins when the user taps the SOS Emergency button. The system displays an emergency warning and asks the user to review the action before continuing. If the user cancels, no emergency alert or message is sent. If the user confirms and selects the trusted-contact flow, the system requests the phone's current location. When location data is available, the coordinates and a map link are attached to the alert; otherwise, the alert continues without location information. The system then creates an emergency record, sends the alert to the primary trusted contact through the server-side delivery service, and displays the delivery result. If the user selects 999, the system opens the phone dialler, but Ethernacare does not automatically place the call.

4.2.2 Virtual Petting Check-In

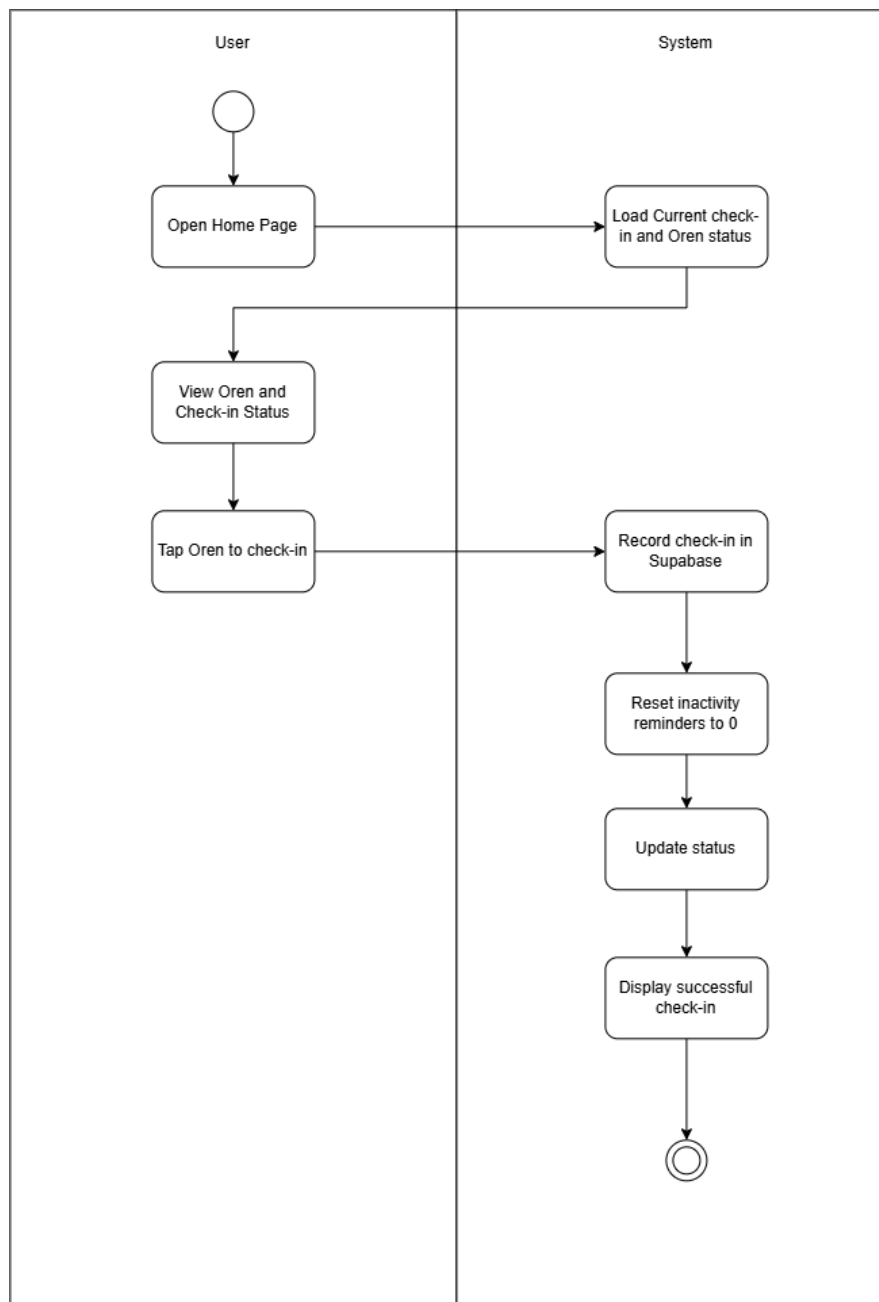


Figure 4.2.2: Activity Diagram show the process of Virtual Petting

The virtual pet check-in activity begins when the user opens the Home page. The system loads the latest check-in record, inactivity reminder count, and Oren's current status from Supabase. The user reviews the displayed information and taps Oren to check in. If the configured check-in window has expired, the system records a new check-in, resets the inactivity reminder count to zero, updates the user and Oren status, and displays a successful check-in message. If the check-in is still current, the system prevents a duplicate record while allowing normal interaction with Oren.

4.2.3 Legacy Planning and Checking Activity

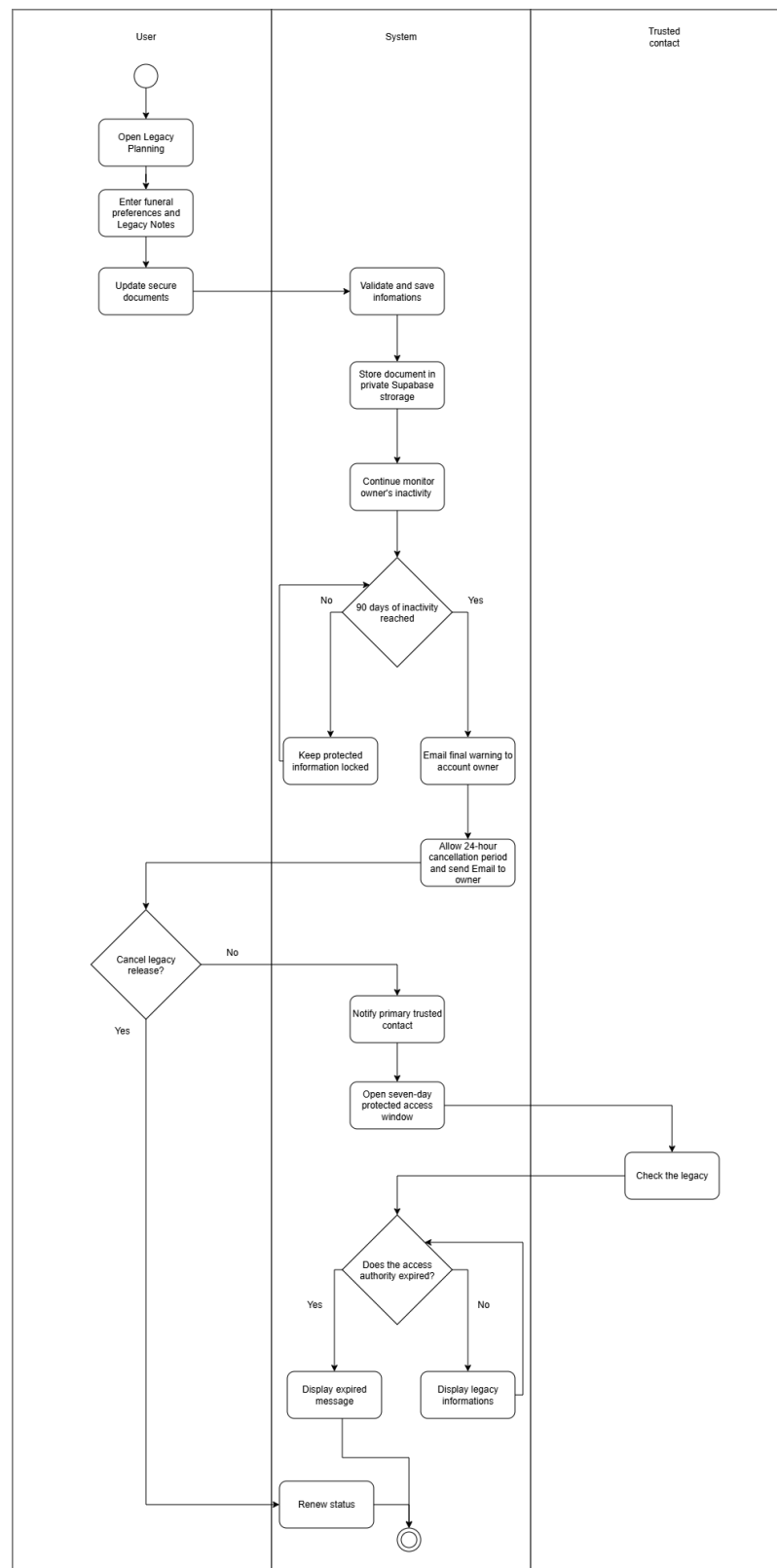


Figure 4.2.3: Activity Diagram show the process of Emergency Alert

The legacy planning activity begins when the user opens the Legacy Planning page and records funeral preferences, Legacy Notes, and secure documents. The system validates the submitted information, rejects sensitive credentials in notes, saves the planning records, and stores uploaded documents in private Supabase Storage. Funeral preferences can be checked by the verified primary trusted contact at any time after verification. Legacy Notes and secure documents remain protected while the system monitors the owner's check-in activity. After 90 days of inactivity, the system emails a final warning to the owner and provides a 24-hour cancellation period. If the owner checks in or cancels the release, the protected information remains locked and the inactivity status is renewed. Otherwise, the system notifies the primary trusted contact and opens a seven-day access window. During this period, the contact must pass the required identity and access checks before viewing the permitted legacy information. After the authority expires, the system rejects access and displays an expiration message.

4.3 ERD Relationship Diagram

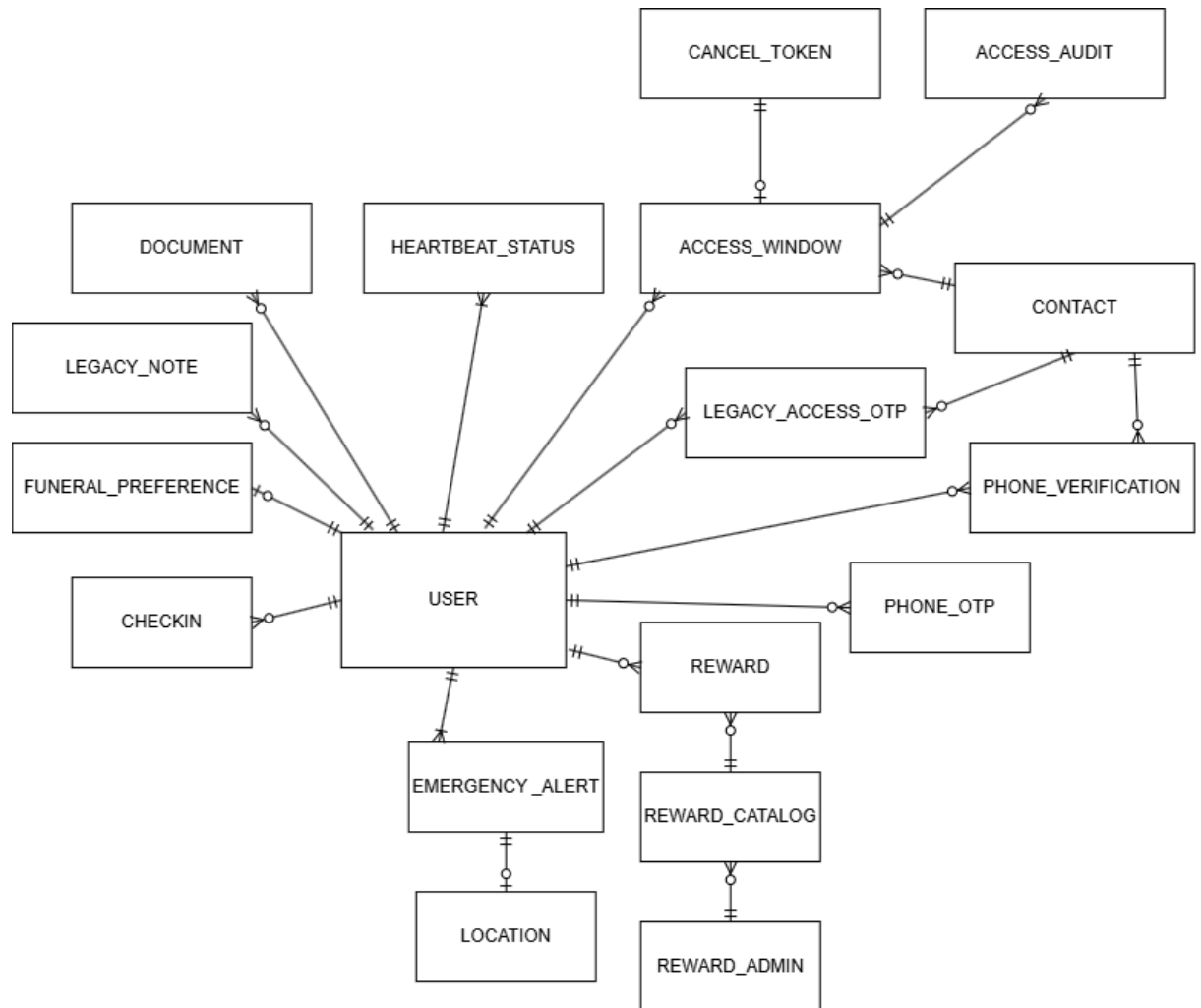


Figure 4.3: ERD diagram that describes the relationship between the entities

4.3.1 Entities attributes

1. User

- Stores the personal information and system settings of the main Ethernacare user.
- Data includes: user ID, name, email, phone number, phone verification time, home address, verified coordinates, blood type, inactivity threshold, escalation target, profile completion time, and legacy-access settings.
- Used to manage authentication, safety monitoring, personal settings, and system interactions. The current system does not store the user's age.

2. Contact

- Stores information about family members or trusted contacts registered by a user.
- Data includes: contact ID, user ID (FK), name, phone number, email, relationship, address, phone verification time, and primary-contact status.
- Used to identify the primary trusted contact who receives emergency and legacy-access notifications.

3. Check-In

- Stores records created when the user checks in by interacting with Oren.
- Data includes: check-in ID, user ID (FK), check-in date and time, and status.
- Used to track user activity, calculate check-in streaks, reset inactivity reminders, and determine whether the user has missed a required check-in period.

4. Inactivity Monitor Status

- Stores the server-calculated inactivity state of each user.
- Data includes: user ID (PK/FK), last check-in time, threshold duration, missed check-in windows, user notification status, escalation time, and last update time.
- Used by the scheduled server worker to identify missed check-ins and trigger the appropriate notification stage.

5. Emergency Alert

- Stores information about SOS emergency events confirmed by the user.
- Data includes: alert ID (PK), user ID (FK), triggered time, and alert status.
- Used to record emergency incidents before notifying the primary trusted contact.

6. Location

- Stores available location information associated with an emergency alert.
- Data includes: location ID (PK), alert ID, latitude, longitude, and timestamp.
- Used to provide coordinates and a map link to the trusted contact when location information is available.

7. Reward

- Stores virtual badges and vouchers earned by individual users.
- Data includes: reward ID (PK), user ID (FK), reward code, streak days, reward type, status, redeem code, earned time, claimed time, and redeemed time.
- Used to maintain the user's reward collection and provide redeem codes for voucher rewards.

8. Reward Catalogue

- Stores the virtual badges and vouchers available in Ethernacare.
- Data includes: reward code (PK), title, sponsor, description, milestone days, reward type, voucher value, catalogue version, and active status.
- Used to define the check-in milestone required to unlock each reward.

9. Reward Admin

- Stores fixed accounts authorised to manage the reward catalogue.
- Data includes: administrator user ID (PK/FK), administrator label, active status, creator ID, and creation time.
- Used to restrict reward creation, modification, activation, and deletion to authorised administrators.

10. Funeral Preference

- Stores the user's funeral and service preferences.
- Data includes: user ID (PK/FK), religion, service type, preferred venue, notes, authorised contact, and update time.
- Used to communicate the user's funeral preferences to the verified primary trusted contact.

11. Legacy Note

- Stores personal wishes and non-sensitive messages created by the user.
- Data includes: note ID (PK), user ID (FK), title, content, creation time, and update time.
- Used to provide permitted legacy information while preventing passwords, PINs, OTPs, and other security credentials from being stored.

12. Document

- Stores metadata for private files uploaded through Legacy Planning.
- Data includes: document ID (PK), user ID (FK), file name, private storage path, and upload time.
- Used to associate uploaded PDF or image files with their owner while the actual files remain in private Supabase Storage.

13. Heartbeat Status

- Stores the user's server-calculated long-term legacy inactivity state.
- Data includes: owner user ID (PK/FK), last heartbeat time, inactive-day count, state, current access-window ID (FK), and update time.
- Used to determine whether protected legacy information should remain locked or enter the legacy-release process after 90 days of inactivity.

14. Access Window

- Stores the temporary period during which a primary trusted contact may access protected legacy information.
- Data includes: window ID (PK), owner user ID (FK), primary contact ID (FK), heartbeat time, state, availability time, expiry time, notification information, and owner cancellation deadline.
- Used to control the final owner-warning period and the seven-day Legacy Checking access period.

15. Cancel Token

- Stores a secure token that allows the account owner to cancel a pending legacy release.
- Data includes: access-window ID (PK/FK), token hash, expiry time, and creation time.
- Used to verify cancellation links without storing the original cancellation token in the database.

16. Access Audit

- Stores audit records when legacy information is released or accessed.
- Data includes: audit ID (PK), owner user ID (FK), contact ID (FK), event type, and creation time.
- Used to maintain a traceable security record of protected legacy-access activities.

17. Legacy Access OTP

- Stores temporary verification codes used during Legacy Checking.
- Data includes: OTP ID (PK), owner user ID (FK), contact ID (FK), phone number, code hash, expiry time, consumed time, attempt count, and creation time.
- Used to verify that the person requesting access controls the registered primary contact phone number.

18. Phone OTP

- Stores temporary codes used to verify user and trusted-contact phone numbers.
- Data includes: OTP ID (PK), user ID (FK), verification purpose, phone number, code hash, expiry time, consumed time, attempt count, and creation time.
- Used to securely perform phone ownership verification without storing the original OTP.

19. Phone Verification

- Stores completed user-phone and contact-phone verification records.
- Data includes: user ID (PK/FK), verification purpose (PK), phone number (PK), and verification time.
- Used to confirm that a phone number was successfully verified before it can be saved or used for safety notifications.

4.3.2 ERD Relationships

User → Check-In (1:M)

One user can perform many check-ins, while each check-in belongs to one user.

User → Contact (1:M)

One user can register multiple trusted contacts. Each contact belongs to one user.

User → Emergency Alert (1:M)

One user can trigger multiple emergency alerts over time.

Emergency Alert → Location (1:0..1)

An alert may contain one location when coordinates are available. The application can continue without location data.

User → Reward (1:M)

One user can earn multiple virtual badges and vouchers.

Reward Catalogue → Reward (1:M)

One catalogue item can be earned by multiple users. Each earned reward identifies one catalogue item through its reward code.

Reward Admin → Reward Catalogue (1:M, management relationship)

An administrator can create, update, activate, deactivate, and delete eligible reward catalogue items. This is an application-level relationship because the catalogue does not store an administrator foreign key.

User → Funeral Preference (1:0..1)

One user may maintain one funeral-preference record.

User → Legacy Note (1:M)

One user can create multiple Legacy Notes.

User → Document (1:M)

One user can upload multiple private legacy documents.

User → Heartbeat Status (1:0..1)

A user may have one current long-term inactivity record.

User → Access Window (1:M)

A user may receive multiple legacy-access windows across different inactivity cycles.

Contact → Access Window (1:M)

A primary trusted contact may be assigned multiple access windows over time.

Access Window → Cancel Token (1:0..1)

A pending access window may have one owner cancellation token.

Access Window → Heartbeat Status (1:0..1)

The heartbeat status may reference the current or latest access window.

User → Legacy Access OTP (1:M)

A user's legacy information may generate multiple OTP requests.

Contact → Legacy Access OTP (1:M)

A primary contact may receive multiple Legacy Checking OTPs over time.

User → Access Audit (1:M)

A user may have multiple legacy-access audit records.

Contact → Access Audit (1:M)

A contact may generate multiple audit records by accessing permitted legacy information.

User → Phone OTP (1:M)

A user may request multiple phone-verification OTPs.

User → Phone Verification (1:M)

A user can have verification records for their own phone and trusted-contact phone numbers.

In the final ERD, keep only one CONTACT entity and connect it to USER, ACCESS_WINDOW, LEGACY_ACCESS_OTP, PHONE_VERIFICATION, and ACCESS_AUDIT.

4.4 Security Handling

4.4.1 Authentication and Access Control (Supabase Authentication)

Ethernacare uses Supabase Authentication to manage account registration, email verification, password recovery, Google authentication, login, and user identity. After successful authentication, Supabase creates a secure JSON Web Token (JWT) session that identifies the user when protected database operations and Edge Functions are requested. Biometric authentication provides an additional device-level application lock, but it does not replace the user's Supabase account credentials. Fingerprint and facial data are validated by the mobile device and are not transmitted to or stored by Ethernacare. Unauthenticated users are prevented from accessing normal application features, while Legacy Checking uses a separate restricted verification process for verified primary trusted contacts.

4.4.2 Data Protection (Supabase Database & Storage)

All user data, including personal details, contacts, check-in records, and documents, are stored securely in the **Supabase database and storage services**. Data access is controlled using Supabase policies, ensuring that users can only access their own data. Sensitive files such as digital wills are stored securely using controlled storage access, preventing unauthorized viewing or modification. Supabase documentation also describes controlled access patterns for private stored assets (Supabase, 2026).

4.4.3 Secure Communication (HTTPS Protocol)

The Flutter application communicates with Supabase, geocoding services, weather services, and other configured APIs using HTTPS. Supabase Edge Functions also communicate with external providers such as Twilio and Brevo through encrypted HTTPS connections. This protects login details, JWT sessions, verification requests, emergency messages, and available location information while they are transmitted over the network.

4.5 UI Prototype

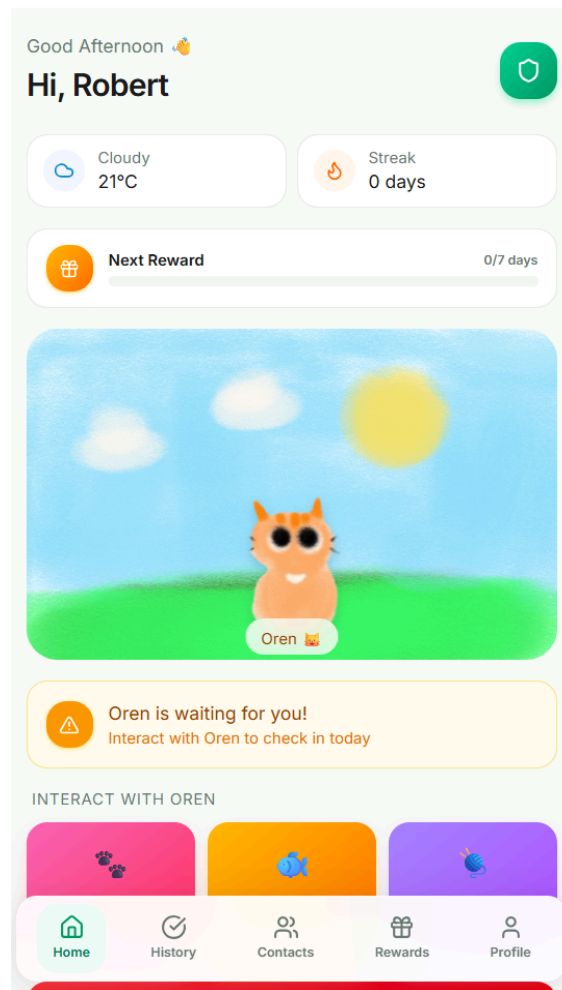


Figure 4.4.1: Home Page

The home page interface is designed for maximum simplicity, featuring a prominent virtual pet dashboard where a friendly cat serves as the central visual focus for the user. To facilitate the daily check-in, the screen includes a large, high-contrast "Pet" button that is easy to identify and tap, accompanied by immediate visual feedback such as floating hearts and cat animations to confirm a successful interaction. Additionally, the UI displays a rewards progress bar that tracks the user's daily streaks toward earning household items, alongside a clear SOS emergency button and status indicators that show the cat's current happiness level and the last recorded check-in time.

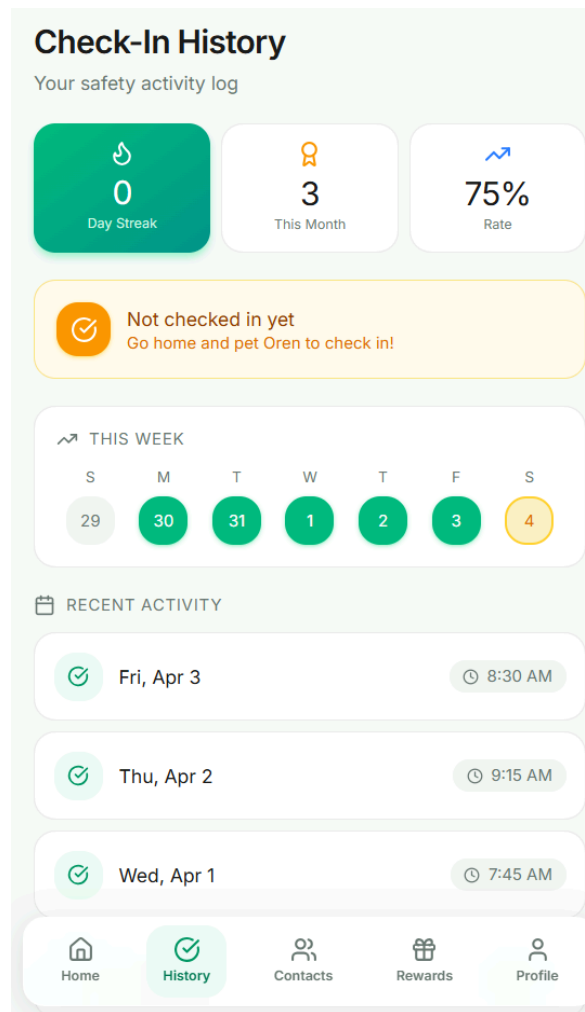


Figure 4.4.2: Check-In History Page

The Check-in History page provides a transparent and structured record of the user's daily interactions, serving as a visual "health diary" of their activity. It presents a chronological list of every successful "petting" event, displaying specific dates and timestamps to confirm when the "heartbeat signal" was sent to the system. This interface allows both the user and their caregivers to monitor consistency, highlighting daily streaks and any missed days that might indicate a change in routine. By centralizing this data, the page reinforces the habit-forming nature of the gamified system while offering peace of mind through a clear, verifiable history of the user's well-being and safety status.

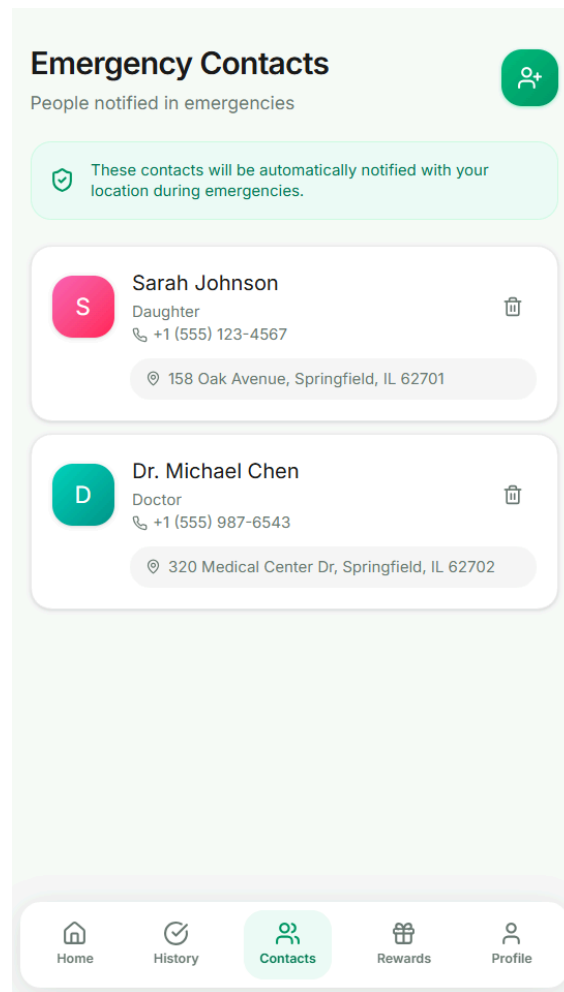


Figure 4.4.3: Emergency Contacts Page

Gemini said The Emergency Contacts page features a clean, card-based layout designed for managing the user's safety network. Each contact is presented on a dedicated card that displays their name, relationship (such as daughter or doctor), phone number, and physical address. A prominent notice at the top informs the user that these individuals will automatically receive location updates during emergencies, while a green "+" button and trash icons allow for the simple addition or removal of contacts. The interface is tied together with a standard bottom navigation bar for quick access to the home dashboard and other app features.

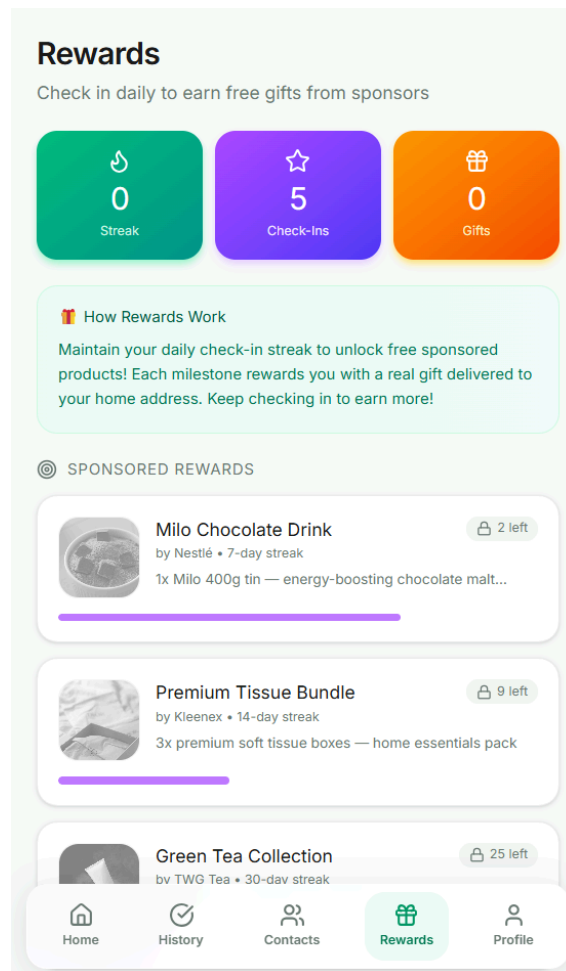


Figure 4.4.4: Rewards Page

The Rewards page serves as the primary incentive hub, motivating users to maintain their daily check-in habits through a tiered gifting system. At the top, three vibrant cards provide a quick summary of the user's current streak, total check-ins, and earned gifts. Below an instructional "How Rewards Work" box, the interface lists available sponsored rewards—such as Milo, tissues, and green tea—each associated with a specific milestone streak (7, 14, or 30 days). Each reward card features a visual progress bar to show how close the user is to claiming the item and a livestock counter to encourage consistent interaction. This layout effectively transforms the safety check-in into a tangible benefit, reinforcing the "nurture-to-reward" model while keeping the navigation simple and goal-oriented.

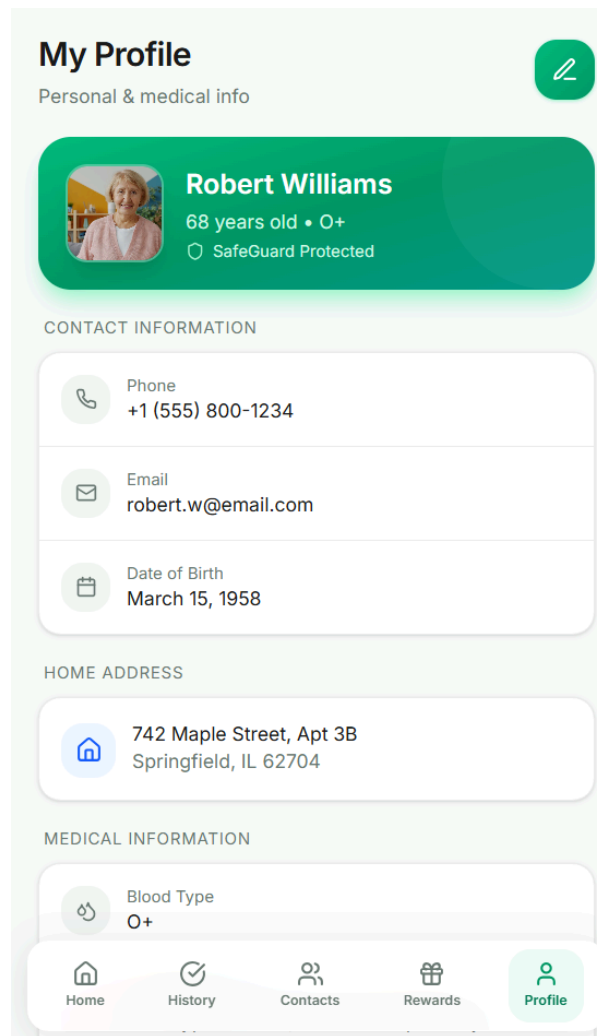


Figure 4.4.5: Profile Page

The My Profile page serves as a centralized hub for the user's personal and health data, designed with high legibility and clear categorization for senior users. The interface features a prominent profile card at the top that displays the user's photo, age, and blood type, alongside a "SafeGuard Protected" badge to provide a sense of security. Below the header, the information is neatly organized into distinct sections for Contact Information, Home Address, and Medical Information, using large icons and high-contrast text to ensure that critical details like phone numbers and addresses are easily readable. This comprehensive layout ensures that essential data is readily available for both the user and emergency responders while maintaining the app's intuitive navigation through the standard bottom menu.

4.6 Chapter Summary and Evaluation

This chapter presented the overall system design of the Ethernacare application, including the architecture, diagrams, and security considerations. The system adopts a three-tier architecture consisting of the Presentation, Business Logic, and Data Access layers to ensure a clear separation of concerns and improve maintainability. Various design models such as the class diagram, entity relationship diagram (ERD), and activity diagrams were developed to illustrate the system structure, data relationships, and process flows, including login and emergency alert functionalities. In addition, appropriate design patterns and security handling techniques were incorporated to enhance system reliability, scalability, and data protection. Overall, this chapter provides a comprehensive blueprint that guides the implementation of the system in the next phase.

Chapter 5

Implementation and Testing

5 Implementation and Testing

This chapter explains how Ethernacare was translated from the requirements and system design into a working safety, well-being, and legacy-planning application. The implementation is organized according to the supplied Chapter 5 guide: system implementation, modules and representative code snippets, sample screens, testing strategies, a module-based test plan, and a final evaluation.

Development was iterative and evidence-driven. User interface changes were checked against compact mobile and wide desktop layouts, while business rules were isolated in services and repositories so they could be tested without a live provider. Higher-risk functions such as inactivity escalation, SMS delivery, phone verification, private documents, Legacy Checking, and emergency location sharing were implemented with explicit states, database controls, and audit-friendly server boundaries.

5.1 System Implementation

Ethernacare uses a layered Flutter-and-Supabase architecture. Flutter widgets form the presentation layer. Controllers and services apply reusable business rules. Repositories isolate Supabase Auth, PostgreSQL, Storage, RPC, and Edge Function calls. Typed models serialize data between layers. On Android, WorkManager performs best-effort local inactivity checks when operating-system constraints permit. Supabase Cron independently invokes the cloud Legacy inactivity worker at 12:00 a.m. Malaysia time.

The application separates local continuity from cloud authority. SharedPreferences stores user-scoped Oren care, weather, dashboard, reward, chat, and reminder state for responsive reopening on the same device. Supabase remains the authority for authentication, profiles, contacts, check-ins, emergency alerts, reward catalogues, earned rewards, Legacy Planning records, private document metadata, OTPs, delivery outboxes, and seven-day Legacy access windows.

5.1.1 Development Tools and Technologies

Technology	Use in Ethernacare	Reason for selection
Flutter 3.38.5 / Dart 3.10.4	Responsive Android, Windows, and web client	One typed codebase with reusable Material widgets and mature test support
Supabase Auth	Email/password, 8-digit email confirmation, password recovery, and configured OAuth	Managed sessions integrated with PostgreSQL user identities
Supabase PostgreSQL	Profiles, contacts, check-ins, rewards, emergency records, OTPs, and Legacy access state	Relational integrity, constraints, triggers, RPCs, and auditable SQL migrations
Row Level Security	Restricts authenticated rows to their owner and protects admin-only reward operations	Database-enforced authorization even when a client request is modified
Supabase Storage	Private PDF/JPEG/PNG Legacy documents up to 10 MB	Private buckets, owner folders, and short-lived signed URLs
Deno / TypeScript Edge Functions	SMS OTP, emergency SMS, AI guidance, Legacy access, cancellation, and daily Legacy worker	Server-side secrets and controlled privileged database operations
Supabase Cron / pg_cron	Runs the production Legacy inactivity worker daily at midnight MYT	Cloud execution continues even when the phone is offline
Python FastAPI reference worker	Readable external-server equivalent in server/python_legacy_server	Demonstrates the same 90-day workflow in a language familiar to the project author

Technology	Use in Ethernacare	Reason for selection
Twilio and Brevo	SMS delivery and transactional email	Provider APIs with server-side credentials and delivery logs
WorkManager and local notifications	Best-effort Android threshold monitoring and local reminder presentation	Supports background work within Android operating-system constraints
data.gov.my and Open-Meteo	Malaysia regional weather with coordinate fallback	Official source first, then a resilient public fallback
Flutter Test / lints / coverage	Logic, service-boundary, widget, responsive, and regression verification	Repeatable evidence before release builds

5.1.2 Modules and Code Snippets

Virtual Pet Check-In and Inactivity Monitoring

The check-in feature records a new check-in only when the user's configured calendar-day period has expired. The server calculates the next valid check-in time using the Malaysia time zone. This prevents duplicate check-ins and ensures that inactivity monitoring follows the selected one-to-seven-day threshold.

SQL:

```
select
max(checkins.checkin_time)

into latest_checkin

from    public.checkins
checkins

where checkins.user_id =
current_user_id;

calculated_next_due := (
(
(latest_checkin at time
zone 'Asia/Kuala_Lumpur')::date
+ configured_days
)::timestamp at time
zone 'Asia/Kuala_Lumpur'
);

if latest_checkin is not
null and now() < calculated_next_due
then

return query

select    false,
latest_checkin,    calculated_next_due,
configured_threshold;

return;
```

The Flutter application calls this protected function through Supabase RPC:

```
Future<bool>
addThresholdCheckin() async {
    final response = await
client.rpc('record_threshold_checkin');

    final row = response is
List && response.isNotEmpty

    ? response.first
    : response;

    if (row is Map &&
row.containsKey('created')) {
        return row['created']
== true;
    }

    throw
    StateError('Supabase did not return the
check-in result.');
```

Explanation:

This code implements the server-controlled check-in process. The database first retrieves the user's most recent check-in using the MAX(checkin_time) operation. This ensures that the check-in decision uses the latest information stored in Supabase instead of relying on the mobile device's local state.

The system calculates calculated_next_due by converting the latest check-in time to the Asia/Kuala_Lumpur time zone, extracting its calendar date, and adding the user's configured number of check-in days. For example, a one-day threshold allows a new check-in after the next midnight, while a two-day threshold requires two calendar days to pass.

If the current time is earlier than the calculated due time, the function returns created = false. This prevents multiple check-in records from being created during the same valid period.

When the check-in is due, the function inserts a new active record into the checkins table and returns the new check-in time.

The application reads the returned created value to determine whether a new check-in was recorded. Keeping this validation in PostgreSQL prevents users from bypassing the check-in period by changing the mobile device's date or modifying the client application.

A successful check-in becomes the user's latest heartbeat. It also causes the inactivity-monitoring cycle to return to its normal state and allows the interface to display the updated check-in and Oren status.

SOS Emergency Alert with Location

When the user confirms an SOS alert, Ethernacare creates an emergency record and requests the current device location. If coordinates are available, they are saved with the alert. The system then creates a server-side delivery record containing the emergency message for the primary trusted contact.

Dart:

```
final alert = await
emergencyRepository.createAlert(
  user.id,
  status: testMode ? 'test_triggered'
: 'triggered',
);

final position = testMode
? null
: await
locationService.getCurrentPosition(
);

final message = testMode
? _testEmergencyMessage()
: _emergencyMessage(position);

final alertId = alert['id'] ??
alert['alert_id'];

if (position != null && alertId != null)
{
  await
emergencyRepository.addLocation(
    alertId: alertId,
    latitude: position.latitude,
    longitude: position.longitude,
  );
}

await
emergencyRepository.createDeliver
yOutbox(
  alertId: alertId,
  userId: user.id,
  contacts: contacts
```

The SOS implementation begins by creating a new emergency record through `emergencyRepository.createAlert()`. The record contains the authenticated user ID, trigger time, and alert status. Recording the alert first ensures that the emergency incident remains in the database even if location retrieval or message delivery later fails.

This requests the phone's current coordinates. In testing mode, location collection is skipped to prevent the test alert from being treated as a genuine location-based emergency.

The system then generates the emergency message. When coordinates are available, the message includes the latitude, longitude, and a map link. The location is also stored in the locations table and associated with the emergency alert ID. If location permission is denied or coordinates are unavailable, the alert continues without location information.

After preparing the message, the application calls `createDeliveryOutbox()`. This creates a delivery record containing the alert ID, user ID, primary trusted contact, and message content. A Supabase Edge Function processes the outbox and sends the message using the configured communication provider.

This design separates emergency recording from message delivery. Temporary provider failures therefore do not remove the original emergency record, and the outbox can retain the delivery status, attempt count, provider response, or error information.

Legacy Planning and Protected Release

The legacy-release worker checks the user's latest heartbeat. When inactivity reaches 90 days, the system prepares an access window and sends a final warning to the account owner. The owner receives one day to cancel the release or complete a new check-in.

SQL:

```
insert into
public.legacy_access_windows (
  owner_user_id,
  heartbeat_at,
  state,
  created_at,
  updated_at
)
select
  snapshots.owner_user_id,
  snapshots.heartbeat_at,
  'owner_notice_pending',
  p_now,
  p_now
from snapshots
where
  snapshots.no_heartbeat_days >=
  90
on conflict (owner_user_id,
  heartbeat_at) do nothing;

update
public.legacy_access_windows
set
  state = 'owner_grace_period',
  owner_notice_sent_at = p_now,
  owner_cancel_deadline = p_now
+ interval '1 day',
  updated_at = p_now
where id = p_window_id;
```

If the owner does not cancel or check in before the deadline, the system opens a seven-day access period for the verified primary trusted contact.

SQL:

```
update
public.legacy_access_windows
set
  state = 'open',
  available_at = p_now,
  expires_at = p_now + interval '7
days',
  notice_sent_at = p_now,
  updated_at = p_now
where id = p_window_id
  and owner_cancelled_at is null
  and owner_cancel_deadline <=
p_now;
```

The Edge Function retrieves eligible records, sends the warning through Brevo, and records whether delivery succeeded:

```
const { data, error } = await
input.supabase.rpc(

"claim_legacy_owner_notice_candi
dates",
  { p_limit: 50, p_now: new
Date().toISOString() },
);

for (const candidate of data ?? []) {
  const delivery = await
sendOwnerWarning({
  apiKey: input.apiKey,
  fromEmail: input.fromEmail,
  fromName: input.fromName,
  supabaseUrl: input.supabaseUrl,
  candidate,
});

  await completeWithRetry({
    supabase: input.supabase,
    rpcName:
"complete_legacy_owner_notice",
    windowId: candidate.window_id,
    success: delivery.ok,
    error: delivery.error,
  });
}
```

The Legacy Planning code implements the protected release of Legacy Notes and secure documents. The server periodically calculates the number of days since each user's latest heartbeat. A heartbeat is normally created by a successful Ethernacare check-in.

When `no_heartbeat_days` reaches 90, the database creates a `legacy_access_windows` record with the state `owner_notice_pending`. The combination of owner user ID and heartbeat time is unique, which prevents the same inactivity cycle from creating multiple access windows.

The system sends a final warning email to the account owner. After successful delivery, the access-window state changes to `owner_grace_period`, and the following value is assigned:

This gives the account owner 24 hours to stop the release. The owner can cancel using the secure cancellation link or complete a new check-in. The cancellation link contains a temporary token, while only its cryptographic hash is stored in the database.

If the owner does not cancel or check in before the deadline, the access window changes to open. The server sets the availability time to the current time and calculates the expiry time using:

The primary trusted contact is then informed that Legacy Checking is available. Access still requires the correct Legacy UID, registered primary-contact phone number, and SMS verification code. After seven days, the access window expires and the protected information becomes unavailable again.

The TypeScript Edge Function coordinates email delivery. It calls `claim_legacy_owner_notice_candidates` to safely claim eligible records in batches of up to 50. For each candidate, it sends the warning using Brevo and calls `completeWithRetry()` to update the database with the delivery result.

This implementation uses a state-based workflow to prevent duplicate emails, provide a final protection period to the owner, restrict access to the primary trusted contact, and automatically close access after the authorised period.

5.1.3 Sample Screens

The following sample screens represent the three main Ethernacare features discussed in the implementation section. The screenshots should be captured from the current mobile build at a consistent portrait resolution, such as 390×844 pixels.

1. Virtual Pet Check-In Screen



Figure 5.1.1: Ethernacare Check-In Screen

Figure 5.1.1 presents the Virtual Pet Check-In screen. The user interacts with Oren to confirm that they are active. After a valid check-in, the system records the activity in Supabase, resets the current inactivity-reminder cycle, updates Oren's status, and awards the applicable tokens. If the current check-in period has not expired, the user may still interact with Oren, but the system does not create a duplicate check-in record.

2. SOS Emergency Screen

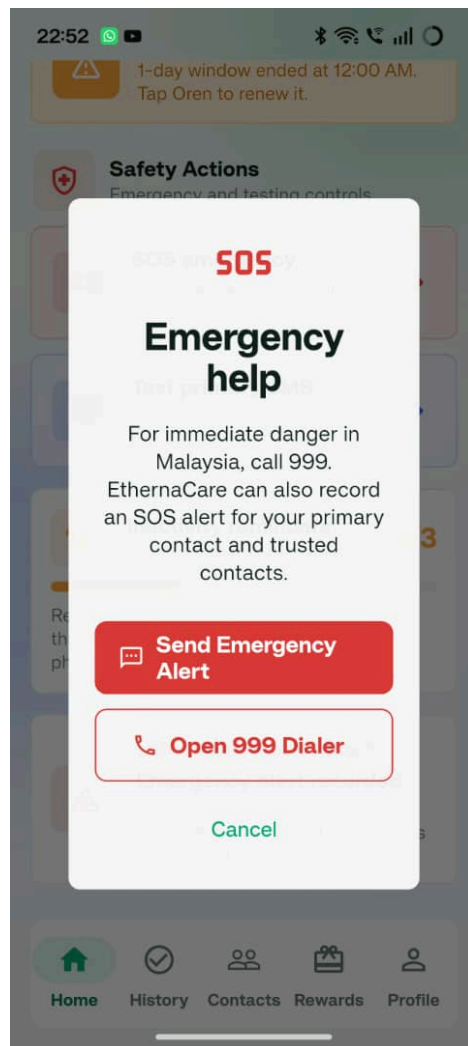


Figure 5.1.2: EthernaCare Emergency Confirmation Screen

Figure 5.1.2 presents the SOS Emergency confirmation screen. The warning prevents an emergency alert from being sent accidentally. The user may cancel the operation, send an alert to the primary trusted contact, or open the Malaysian 999 dialler. When an alert is confirmed, EthernaCare records the emergency event and attempts to obtain the current location. Available coordinates and a map link are included in the trusted-contact message, while the alert can continue without location when permission or coordinates are unavailable.

3. Legacy Planning and Checking Screen

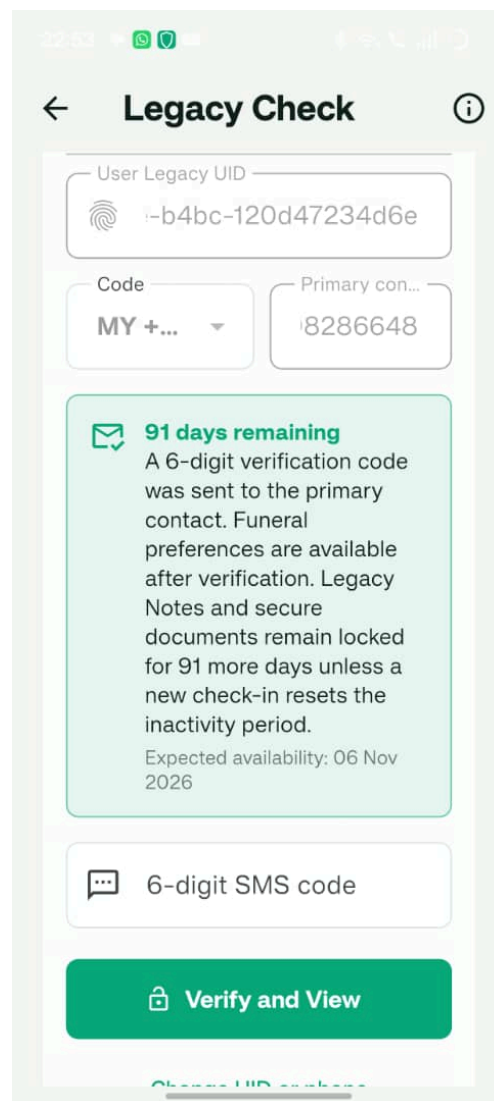


Figure 5.1.3: Ethernacare Emergency Confirmation Screen

Figure 5.1.3 presents the Legacy Checking screen used by the primary trusted contact. The contact must provide the account owner's Legacy UID and the registered primary-contact phone number before requesting access. Funeral preferences may be viewed after successful verification, while Legacy Notes and secure documents remain protected until the 90-day inactivity requirement, 24-hour owner-protection period, and access checks are satisfied. When access is approved, the system displays the permitted information only during the seven-day access window.

5.2 Testing Methods

5.2.1 Functional Testing

Unit and logic testing validates threshold boundaries, escalation rules, Oren energy decay, reward calculations, validation, serialization, Legacy content filtering, and provider fallback decisions.

- Service-boundary testing uses mocked repositories and Supabase responses to verify orchestration without changing production records.
- Widget testing renders Oren, contacts, rewards, Legacy, dialogs, and guidance states at controlled viewports to detect unreachable controls and overflow.
- Integration checks verify deployed Edge Function health, migration parity, private-storage expectations, release builds, and platform manifests.
- Context-based testing describes the real situation of a user, owner, primary contact, or reward administrator instead of testing an isolated button without purpose.

5.2.2 Non-Functional Testing

Reliability testing checks duplicate suppression, one-time escalation, retry visibility, cache restoration, and fallback behavior. Security testing checks RLS ownership, recent phone verification, private Storage paths, signed URLs, OTP expiry and attempts, Legacy audit events, admin authorization, and server-secret boundaries. Usability checks cover scrolling, safe areas, readable status text, guidance, full-page forms, and mobile viewport behavior. Maintainability is supported by repositories, services, typed models, idempotent migrations, and separate Edge Functions.

5.3 Context-Based Test Plan and Test Cases

The cases below use realistic contexts with free user input rather than one fixed data set. For each run, the participant or tester chooses or randomly generates valid, boundary and invalid values within the stated field rules. Names, email addresses, phone numbers, thresholds, dates, times, locations, notes, files, catalogue values and device settings should vary between runs. The tester records the actual values, device, time and generated seed (when used) so a failure can be reproduced without turning the test case into a scripted demonstration.

5.3.1 Test Cases

Table 5.3.1: Check-in, inactivity, and emergency

ID	Context and Action	Expected Outcome	Evidence / Status
SAFE-01	Request a check-in before, at, and after the configured threshold.	A new check-in is stored only when the server reports that it is due.	Automated service tests PASS
SAFE-02	Evaluate one, two, and three missed threshold windows and repeat the worker calculation.	The missed-window stage follows elapsed time. SQL delivery keys are designed to prevent duplicates.	Threshold calculations PASS
SAFE-03	Process reminders using verified and unverified recipients with successful and failed provider responses.	Production SMS is queued server-side and restricted to approved recipients and message categories.	Mocked service tests PASS

ID	Context and Action	Expected Outcome	Evidence / Status
SAFE-04	Open SOS and select Cancel, Send Emergency Alert, or Open 999 Dialer. Allow or deny location.	Cancel performs no action. Alert confirmation attempts location and queues an alert. The dialler opens only as an external application and Ethernacare cannot place the call.	Service tests PASS
SAFE-05	Background, close, or resume the application while inactivity time passes.	The phone may show a local reminder when evaluated. The scheduled server worker operates independently every 15 minutes.	Local orchestration tests PASS

Table 5.3.1.2: Trusted-contact and phone-verification

ID	Context and Action	Expected Outcome	Evidence / Status
CONTACT-01	Enter a valid name, relationship, phone, email, and Malaysian address, then save.	The contact is submitted for the signed-in user and the contact cache is refreshed.	Form and mocked service tests PASS
CONTACT-02	Request and enter valid, invalid, expired, or incorrect phone OTP values.	Only a current OTP associated with the correct user, purpose, and phone can verify the number.	Phone validation tests PASS
CONTACT-03	Add contacts up to the limit, attempt a duplicate, and change the primary contact.	Maximum five contacts, no duplicate normalized phone numbers, and at most one primary contact.	Validator and service tests PASS
CONTACT-04	Attempt to access another account's contact using a different authenticated user.	Supabase RLS should reject the cross-user request.	Pending integration; no live two-account RLS test has been completed.

Table 5.3.1.3: Oren, rewards, and admin

ID	Context and Action	Expected Outcome	Evidence / Status
OREN-01	Evaluate Oren after whole hours, partial hours, future timestamps, and zero energy.	Energy decreases once per complete hour, preserves partial time, and never falls below zero.	Automated tests PASS .
OREN-02	Attempt Play below, at, and above the required energy.	Play succeeds only with sufficient energy; rejected attempts consume nothing.	Automated service tests PASS
OREN-03	Buy affordable and unaffordable items, select an owned item, play, and reload.	Valid purchases deduct tokens once; invalid purchases and unowned selections are rejected.	Service and cache tests PASS
REWARD-01	Reach a badge milestone and attempt to collect the badge repeatedly.	An eligible badge enters the Reward Collection once.	Service and model tests PASS
REWARD-02	Reach a voucher milestone and open Check Reward across refreshes.	The earned voucher and its redeem code remain available in the Reward Collection.	Service and widget tests PASS
ADMIN-01	Create or edit a reward using valid and invalid values, then refresh.	Validation rejects incomplete values and saves valid catalogue entries.	Editor widget tests PASS
ADMIN-02	Select and delete a catalogue reward.	The admin interface requests deletion; database	Implemented, but automated deletion and

		references determine whether deletion succeeds.	live database tests are pending.
--	--	---	----------------------------------

Table 5.3.1.4: Legacy Planning and Legacy Checking

ID	Context and Action	Expected Outcome	Evidence / Status
LEGACY-01	Save funeral preferences, a safe note, and a supported file no larger than 10 MB.	Planning values are saved and documents undergo extension, size, signature, and ownership validation.	Service and validation tests PASS
LEGACY-02	Enter ordinary text and password-, PIN-, OTP-, or recovery-secret-like notes.	Ordinary notes are accepted; recognized credential-like content is rejected.	Automated validation tests PASS
LEGACY-03	Evaluate the 90-day threshold, 24-hour protection period, cancellation, seven-day window, and expiry.	Server code defines warning, cancellation, release, and expiry states.	Flutter parsing tests PASS
LEGACY-04	Submit valid and invalid Legacy UUIDs, phones, and verification codes.	Only the verified primary contact with a current code can proceed without exposing protected content.	Response and UI tests PASS
LEGACY-05	Access preferences, notes, and documents inside and outside the release window.	Preferences support verified access; notes and documents require an active protected-content window.	Result parsing tests PASS

5.4 Chapter Summary and Evaluation

This chapter documented the current Ethernacare implementation according to the supplied guide. The application combines a layered Flutter client with Supabase authentication, PostgreSQL, RLS, private Storage, Realtime, Edge Functions, and a scheduled Legacy worker. The key business rules are now explicit: check-ins follow the configured rolling threshold; the first missed window triggers a local reminder; the second triggers a user SMS attempt; the third escalates once to the trusted contact; Oren energy decays hourly; virtual badges and vouchers have distinct redemption behavior; and Legacy protected content follows a 90-day, 24-hour owner-protection, and seven-day contact-access sequence

Chapter 6

Discussions and Conclusion

6 Discussions and Conclusion

Summary

EthernaCare was developed to reduce delayed awareness when an older adult or independent individual living alone stops providing an expected safety signal. The solution combines a configurable Oren check-in, staged inactivity reminders, verified trusted contacts, manual SOS support, location-assisted alerts, virtual rewards, weather-responsive interaction, AI guidance and consent-controlled Legacy Planning. Flutter supports a consistent multi-platform client, while Supabase provides managed identity, relational data, security policies, private files and scheduled server execution.

Achievements

The project achieved its principal functional objectives. Check-ins now follow a one-to-168-hour rolling threshold instead of a fixed daily flag. A cloud worker evaluates every 15 minutes and persists the first, second and third missed-window stages. Contact phone verification, duplicate suppression, bounded SMS requests, stale-job recovery and server delivery state support safer escalation. Oren has hourly energy decay, user-scoped care state and serialized mutations that prevent lost rapid-tap updates. Dashboard refreshes reject stale overlapping responses and refresh on app resume. Rewards are fully virtual, with badge collection, voucher codes and a separate catalogue-admin route. Legacy Planning implements owner consent, funeral preferences, secret-aware notes, private documents, a 90-day trigger, a 24-hour owner-protection step and seven-day protected access. Supported sessions can be locked by asynchronously coordinated device biometrics, while the first-run guide and Settings module improve onboarding and accessibility.

Quality evidence includes 138 passing automated tests and 30.64 percent line coverage. The tests exercise important service, repository and widget paths, including threshold calculations, Oren decay and concurrency, OTP rules, Legacy timing, virtual rewards, guide/settings persistence and biometric prompt behaviour. This gives repeatable regression evidence, although it does not replace physical and provider acceptance testing.

Contributions

The main contribution is the integration of a low-friction virtual companion with an explicit, configurable safety heartbeat and a server-controlled escalation model. Separating care actions from the heartbeat reduces ambiguity, while virtual rewards support engagement without physical delivery administration. The Legacy design also contributes to a consent-based staged release: the owner receives a final protection opportunity before time-limited access is offered to a verified primary contact.

Limitations and Future Improvements

EthernaCare is not a medical device and cannot detect falls, unconsciousness, vital signs, death or the cause of inactivity. It does not automatically dispatch Malaysian emergency services. SMS and email delivery depend on provider accounts, carrier filtering and valid contact data. Android local scheduling is best-effort, GPS depends on permission and signal, and biometric authentication protects a restored local session rather than proving the identity of a trusted contact.

Future work should raise automated coverage, add provider sandbox and delivery-status integration tests, complete a structured physical-device matrix, add accessibility testing with older adults, introduce operational dashboards and alerting, support secure cross-device synchronization for Oren state, and perform professional privacy, legal and security reviews. Before production release, sensitive offline profile or medical caches should be removed when unnecessary or moved from unencrypted preferences to protected platform storage (OWASP Foundation, n.d.). The Android application identifier should also be changed from the development identifier, and direct SEND_SMS permission should be retained only after confirming Google Play eligibility and declaration requirements; otherwise, server-side SMS should remain the primary path (Google Play, n.d.). A production release should also define data retention, incident response, account recovery and support procedures.

Issues and Solutions

A major issue was that phone-only background execution could not guarantee threshold processing after the app was closed. This was addressed by moving authoritative evaluation to a 15-minute Supabase Cron worker while retaining WorkManager for local reminders. Duplicate SMS risk was reduced through persisted stages, idempotency keys and delivery outboxes; provider hangs and interrupted work were addressed with a 12-second timeout, retry states and five-minute stale-job recovery. Rapid Oren actions that could overwrite local state were serialized, and dashboard requests now discard stale overlapping results and refresh on app resumes. OAuth users missing public profile rows were addressed with profile bootstrap logic. Private Legacy files were protected with RLS, private Storage and short-lived signed URLs. Repeated biometric prompts were addressed through asynchronous request coalescing and lifecycle-aware gating.

External delivery and platform behaviour remain the most important unresolved acceptance risks. The project therefore distinguishes a successful provider request from confirmed receipt and labels physical SMS, email, GPS, Android lifecycle and biometric-sensor tests separately from automated evidence.

Conclusion

The completed prototype demonstrates that a friendly check-in experience can be combined with explicit safety rules, server scheduling and privacy-conscious Legacy controls. Ethernacare provides useful support and earlier notification, but it must be presented honestly as a supplementary well-being system rather than a guaranteed emergency or medical service. Within those boundaries, the project satisfies its main academic objectives and provides a maintainable foundation for further validation and development.

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Appendices

In order to enhance better understanding of the project, students should as far as possible include all directly relevant materials, figures or diagrams in the **main body** rather than in the Appendix. The appendix is reserved only for items which may not directly be relevant or essential to enhance a reader's understanding of the project, or which may interrupt the smooth reading of the project document (for example being too voluminous).

Appendices should only include supportive materials **directly referred** to in the writing and should be kept to a **minimum**, e.g. selected pages of an annual report, not the entire document. Examples of items included in Appendices are:

- Company's report and documentation, such as sample invoice, purchase order form, etc.
- Project meeting documentation e.g. minutes of meetings, tracking documents, memos etc.
- Questionnaires and results, interview questions and results, pilot test and results, observation sheet and results, experiment test plan and results, etc.
- Analysis/design diagrams (only those not incorporated in the main body of the report).

If there is more than one appendix, they should be identified as A, B, etc (e.g. Appendix A). Formulae and equations in appendices should be given separate numbering: Eq. (A.1), Eq. (A.2), etc.; in a subsequent appendix, Eq. (B.1) and so on. Similarly for tables and figures: Table A.1; Fig. A.1, etc.

IMPORTANT NOTE TO STUDENTS

APPENDIX *n* User Guide

- List the username and password of multiple roles (if applicable)
- Provide clear screen shots of each page, explain the functions of each button

As a rough guide, the user guide should include the following sections:

System Document

In this section, students should provide the following pieces of information:

- **System (hardware and software requirements).** Students should describe the minimum hardware and software requirements to install the software application which has been developed by the students, for example, DBMS, OS, program development tools etc.
- **Installation.** Under this section, students should create an ‘installation’ CD and provide a brief step-by-step guide on how a new user can install the software on a computer system. Students should indicate any special setup information, such as the specific location of placing the database files, the SQL statements to add tables, etc. Software source code should also be included in this CD. Students are not required to print out the software code.

Operation Document

Under this section, students are required to provide a brief step-by-step guide on how to use the installed software. The guides should teach the user how to run the software and use its major functions and features. For example, steps show guide the users on how to run the system, e.g. to use an executable file or to use the IDE. The login information such as username and password for each of the different users (or roles) must be provided. Some screen interfaces would be useful.

APPENDIX $n+1$ Developer Guide*

*To be included for **Smart Campus Projects** and **Real-Life Projects**. This section should include the following:

- List the necessary software, installer, API, library that must be installed
- List the authentication details, such as username and password for all security

Other Appendices:

- Supporting documents
- Non-disclosure agreement (if applicable)
- Other detailed documentation, such as important references, interview results, survey results, etc.

Appendix A: System installation

1. Minimum End-User Hardware

Android device

- Smartphone with at least 2 GB RAM and 250 MB free storage.
- Internet connection for authentication, database access, SMS processing, weather, rewards, and Legacy Checking.
- Android notification support.
- GPS is required to attach current location to SOS alerts.
- Fingerprint or facial-recognition hardware is optional.
- Mobile service is required to receive verification SMS messages.

Windows computer

- 64-bit processor with at least 4 GB RAM.
- At least 500 MB free storage.
- Internet connection.
- Windows Hello-compatible hardware is optional for biometric authentication.

2. Minimum Supported Software and Operating Systems

- Android 7.0 or later, API level 24 or higher.
- Windows 10 or Windows 11, 64-bit.
- A valid email account for registration and password recovery.
- Notifications and location permissions should be enabled on Android.
- Ethernacare operates in portrait orientation on mobile devices.
- No local DBMS installation is required. Supabase Postgres is the primary database.
- SharedPreferences stores limited device settings and cached data; Ethernacare does not use SQLite.

3. Developer and Backend Requirements

- Flutter 3.38.5.
- Dart 3.10.4.
- Android Studio with Android SDK and Java 17 or later.
- Visual Studio 2022 with the **Desktop development with C++** workload for Windows builds.
- Git, Node.js/npm, and the Supabase CLI.
- Supabase services: Authentication, Postgres, Storage, Realtime, Edge Functions, Vault, and Cron.
- Twilio credentials for phone verification and SMS alerts.
- Brevo or another SMTP provider for account verification and password-recovery emails.
- Brevo API credentials for Legacy inactivity emails.
- Gemini or another configured AI provider for online AI Guidance.
- The supplied Python worker is an alternative reference implementation. Production currently uses Supabase Edge Functions.

4. Installation Package/CD Contents

The submission package should contain:

/Android/EthernaCare.apk

/Windows/EthernaCare/ – (Entire Windows release folder)

/Source/ – (Flutter source code)

/Source/supabase/migrations/ – (Database migrations)

/Source/supabase/functions/ – (Edge Function source code)

/Documentation/ – (System and operation documents)

/TestEvidence/ – (Test results and screenshots)

/README.txt – (Installation summary)

Do not include Supabase service-role keys, Twilio tokens, SMTP passwords, API keys, or other secrets.

5. Android Installation

1. Copy Ethernacare.apk to the Android device.
2. Open the APK using the device's file manager.
3. Allow installation from the selected source when requested.
4. Select Install, then open Ethernacare.
5. Grant notification and location permissions.
6. Optionally enable biometric unlocking under Profile > Biometric unlock.

Developers can generate the APK with:

1. `flutter pub get`
2. `flutter build apk --release`

The APK is produced at:

`"build/app/outputs/flutter-apk/app-release.apk"`

The current release build uses development signing and must receive production signing before Play Store distribution.

6. Windows Installation

1. Extract the complete Ethernacare Windows folder.
2. Keep the executable, DLLs, and data folder together.
3. Run Ethernacare.exe.
4. Allow network access if Windows Firewall requests it.

Build the Windows package with:

1. `flutter pub get`
2. `flutter build windows --release`
3. Distribute the complete folder:
4. `build\windows\x64\runner\Release\`

7. Supabase Database and Edge Function Setup

From the Flutter project directory:

1. `npx --yes supabase login`
2. `npx --yes supabase link --project-ref mekiduxpnrorkfphjgpc`
3. `npx --yes supabase db push`

For a new Supabase project, the base SQL scripts under `supabase/` must also be applied through the SQL Editor. These cover profiles, contacts, phone OTP, alerts, Legacy Planning, document storage, notes, RLS, and validation. Files named `quick_fix_*` are intended for repairing older deployments.

Deploy the production Edge Functions:

1. `npx --yes supabase functions deploy request-phone-otp --project-ref mekiduxpnrorkfphjgpc`
2. `npx --yes supabase functions deploy verify-phone-otp --project-ref mekiduxpnrorkfphjgpc`
3. `npx --yes supabase functions deploy send-emergency-sms --project-ref mekiduxpnrorkfphjgpc`
4. `npx --yes supabase functions deploy process-inactivity-thresholds --project-ref mekiduxpnrorkfphjgpc`
5. `npx --yes supabase functions deploy request-legacy-access --project-ref mekiduxpnrorkfphjgpc`
6. `npx --yes supabase functions deploy verify-legacy-access --project-ref mekiduxpnrorkfphjgpc`

7. `npx --yes supabase functions deploy process-legacy-inactivity --project-ref mekiduxpnrorkfphjgpc`
8. `npx --yes supabase functions deploy cancel-legacy-release --project-ref mekiduxpnrorkfphjgpc`
9. `npx --yes supabase functions deploy finalize-legacy-document --project-ref mekiduxpnrorkfphjgpc`
10. `npx --yes supabase functions deploy ai-guidance --project-ref mekiduxpnrorkfphjgpc`

Required secrets include:

- `PHONE_OTP_SECRET`
- `TWILIO_ACCOUNT_SID`
- `TWILIO_AUTH_TOKEN`
- `TWILIO_FROM_NUMBER`
- `LEGACY_CRON_SECRET`
- `BREVO_API_KEY`
- `LEGACY_NOTICE_FROM_EMAIL`
- `LEGACY_NOTICE_FROM_NAME`
- `GEMINI_API_KEY`, when online AI Guidance is enabled

Run `supabase/legacy_inactivity_cron_setup.sql` after replacing its placeholders. The inactivity-threshold worker runs every 15 minutes, while the 90-day Legacy worker runs daily at midnight Malaysia time.

8. Verification After Installation

Run the automated checks:

1. flutter analyze
2. flutter test
3. flutter build apk --release
4. flutter build windows --release
5. npx --yes supabase migration list --linked
6. npx --yes supabase functions list --project-ref mekiduxpnrorkfphjgpc

Physical-device verification should cover registration email, phone OTP, biometrics, notifications, location, SOS delivery, background inactivity processing, and Legacy document access.

Appendix B: User Guide

B.1 Account and First-Time Setup

Use a personal email address or a configured Google account, accept the current Terms and Conditions, complete the required profile, verify the user's phone and create a verified primary trusted contact. The first completed setup opens an eight-step guide to the main screens. It may be skipped and replayed later from Profile or Settings. Never share a password or verification code, and do not use a shared demonstration password.

B.2 Daily Check-In and Safety Monitor

Open Home and review Oren's status and the Safety Monitor. Tap Oren only when the rolling threshold allows the safety check-in. Feed and Play affect Oren's care state but do not count as a heartbeat. History shows successful check-in timestamps. If the status becomes overdue, follow the displayed guidance and check in as soon as it is safe to do so.

B.4 Manual SOS and Emergency Boundaries

Use SOS only when a verified trusted contact should be notified. Location is attached only when permission is granted and coordinates are available. Review the confirmation before sending. The 999 action opens an explicit call flow and should be used only for a real emergency; Ethernacare does not automatically dispatch emergency services and does not replace direct emergency contact.

B.3 Trusted Contacts

Add only people who have agreed to act as trusted contacts. Enter a valid international phone number, complete SMS verification and select one primary contact. Review the primary contact whenever a number changes. A trusted contact is an emergency and Legacy recipient; the contact does not receive the owner's full application account or unrestricted data access.

B..5 Oren, Shop and Virtual Rewards

Feed and Play change Oren's energy and mood. Tokens purchase virtual toys and care items. Check-in milestones may unlock virtual badges or vouchers; collected badges appear in My Badge List and an earned voucher may show a stable user-specific code. These rewards are engagement features and do not change the emergency threshold.

B.6 Weather and AI Guidance

Weather uses the saved Malaysian state and region first, with location-based or public-service fallback where available. AI Guidance may answer general questions about elderly care, emergency preparation or funeral planning. Treat the response as general information only. Do not use it for diagnosis, medication, legal advice or an emergency; contact an appropriate professional or emergency service when needed.

B.7 Legacy Planning and Contact Access

The owner may record funeral preferences, create protected Legacy Notes and upload supported private documents. Select the existing verified primary contact and enable release consent only after reviewing the explanation. Never store passwords, PINs, OTPs, recovery phrases or account credentials in a Legacy Note. Funeral preferences and protected content follow different access rules; notes and documents remain time-restricted and require the staged owner-protection process.

For a contact check, open the public Legacy flow, enter the owner's Legacy UID and complete the SMS verification issued to the authorised phone. Do not forward the code. If access is not yet eligible or has expired, the system should show a safe status without revealing protected notes or documents.

B.8 Profile, Accessibility and Privacy

Use Profile to review contact details, address, blood type, rolling threshold, escalation preference, phone verification, biometric session protection and Legacy UID. In Settings, choose reminders, Oren sounds, text size and reduced motion, replay the guide or restore defaults. Sign out on a shared device. Biometric protection secures the restored local session on supported hardware; it does not replace the account password or server authentication.

B.9 Troubleshooting

If a check-in is unavailable, confirm that the selected rolling threshold has elapsed and refresh History. If an OTP does not arrive, verify the number and network, wait for the current request to finish and use the supported resend flow; never guess or reuse an old code. If weather or AI is unavailable, retry after connectivity returns; safety check-in and direct contact actions should remain the priority. If location is missing from SOS, review device permission and positioning. If a file will not upload, confirm that it is a supported PDF, JPEG or PNG within the displayed size limit.

B.10 Safety and Privacy Checklist

Keep one primary contact verified and current.

Choose a threshold that the user understands and can realistically meet.

Use synthetic personal data during demonstrations and testing.

Do not place credentials or recovery secrets in Legacy Notes or report screenshots.

Confirm real SMS, email, GPS, background and biometric behaviour on the intended physical device before relying on a release.

Use direct emergency services for an actual emergency; Ethernacare is a safety-support tool, not a medical device or guaranteed dispatch service.

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